Hysteresis Indicies and Beta of IBW sites: Curry, Silverado, and Lake Margarita

2026-03-26

Aims to plot hysteresis per analyte per storm per site

Metrics

- *Hysteresis index* is calculated as the median (+/- 95% CI) across 2% intervals of discharge.

  +HI = clockwise (source limitation or nearstream source)

  -HI = counterclockwise (transport limitation or distal source)

- *Beta* is the slope of the CQ relationship on the rising limb of the storm

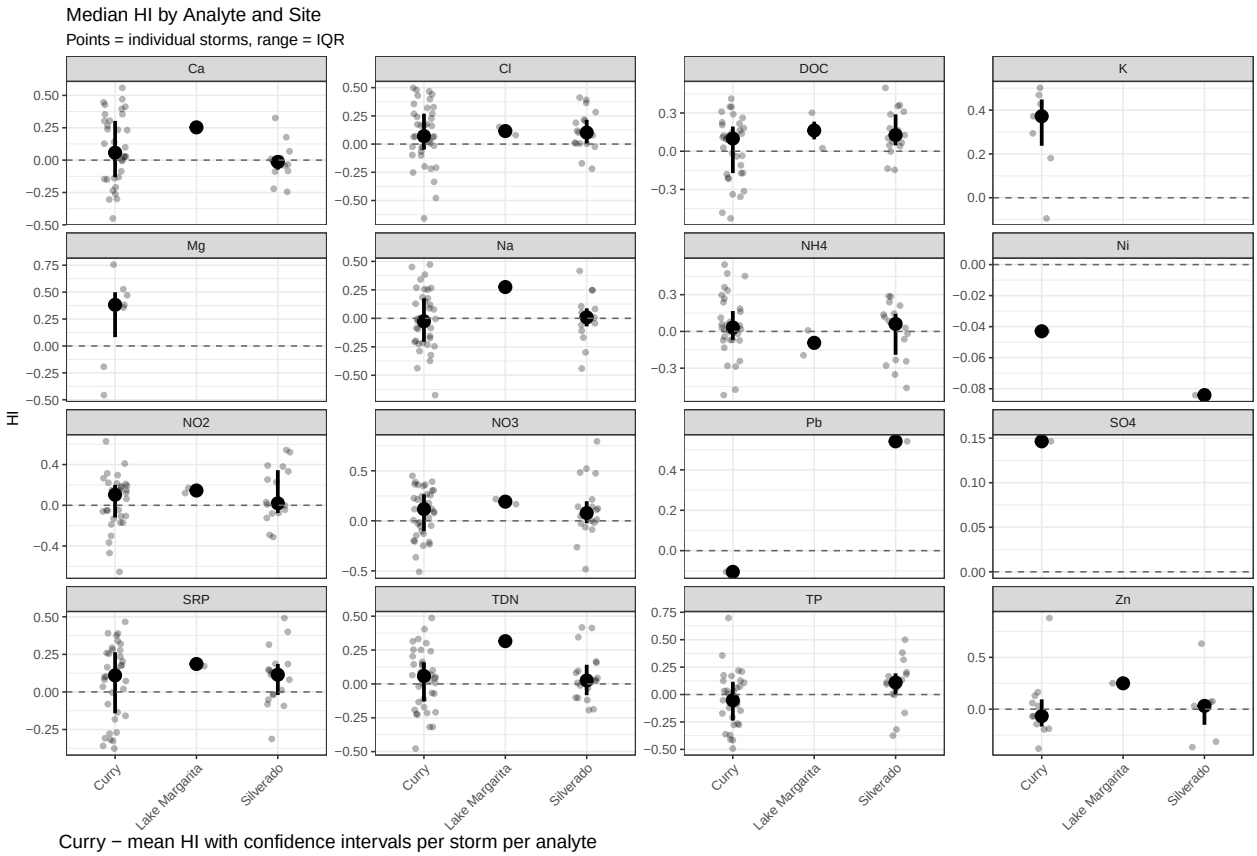
  CIs on beta are bootstrapped 95% CI

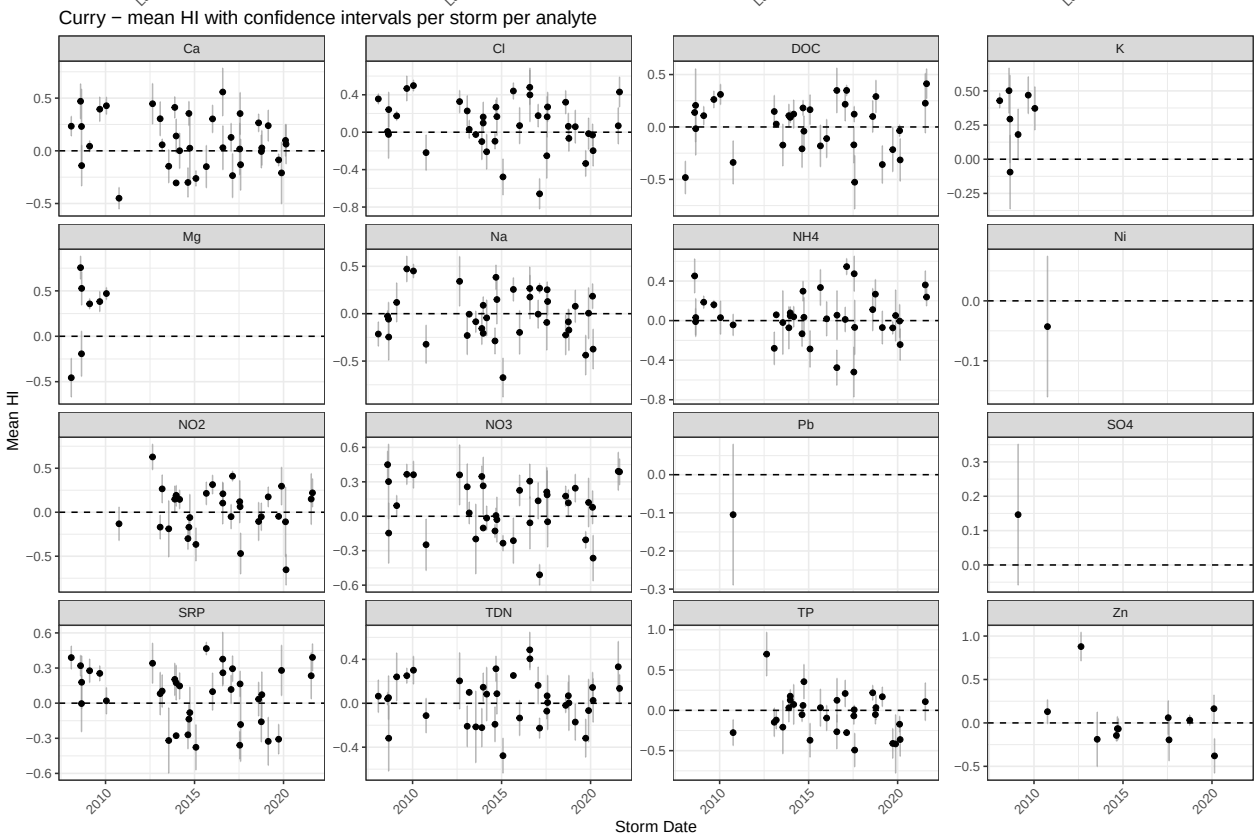
Data munging/cleaning

Workflow before this rmd, in order:

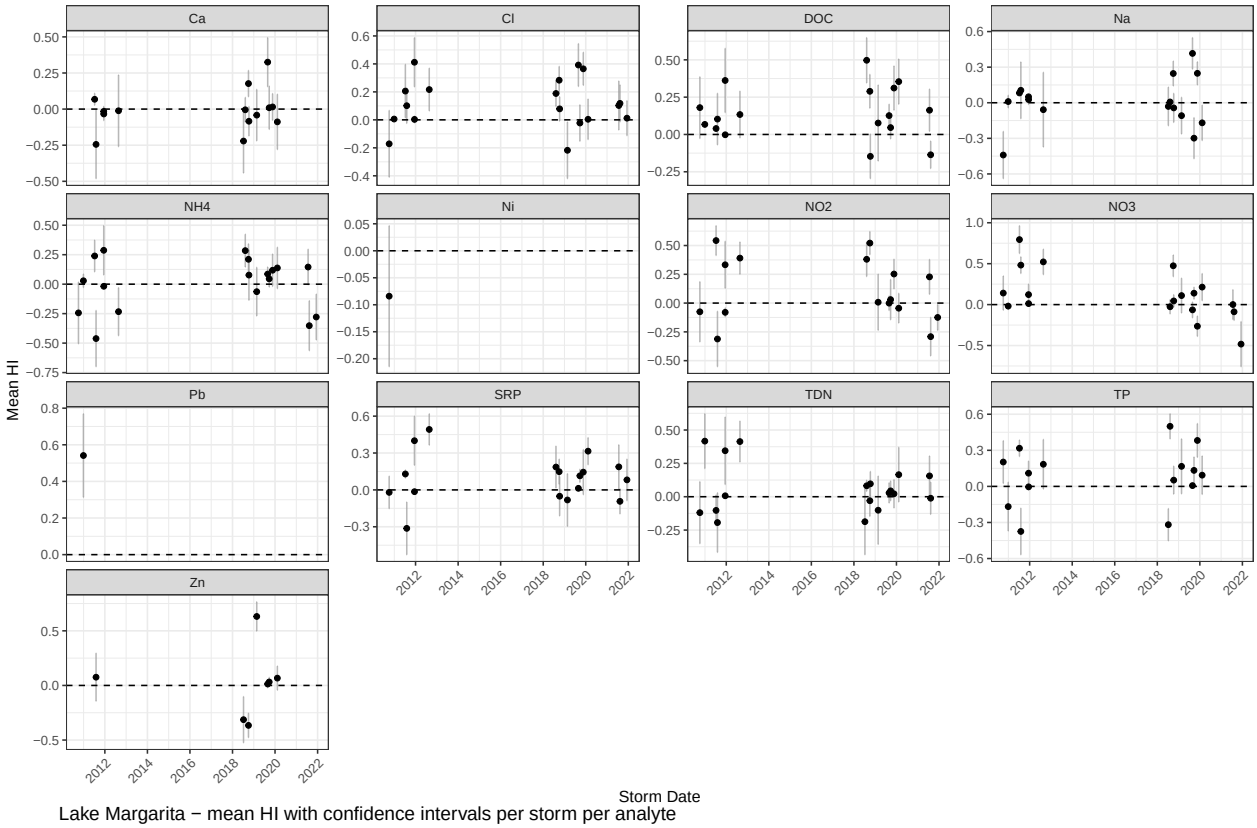
- q_storms.R: import discharge from Lake Margherite, Silverado, & Curry, delineate unique storms, and interpolate Q
- merge_Q_chems.R: merge discharge data with chem data per site, produce brief cq plots per analyte per site
- hysteresis_calc_2.R: calculate hysteresis based on python function, removed a lot of storms with not enough data

Plotting Hysteresis

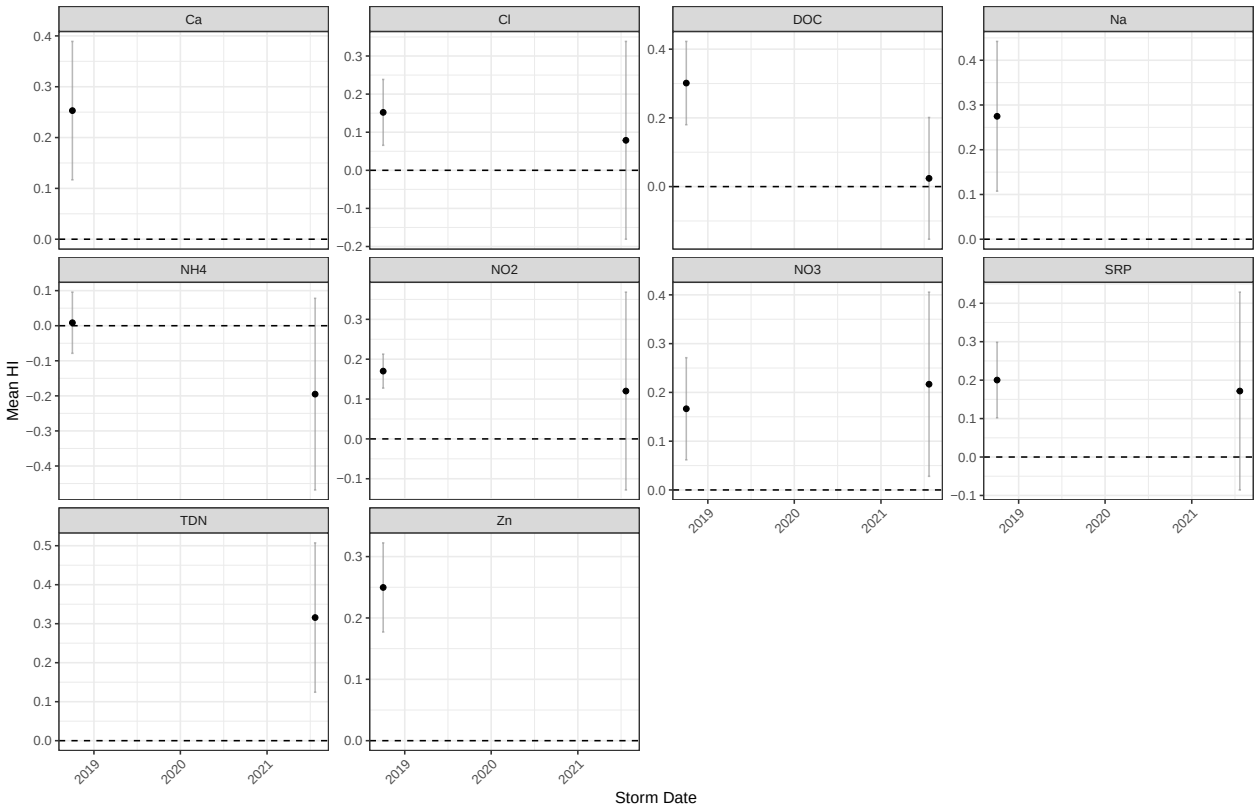




Silverado − mean HI with confidence intervals per storm per analyte

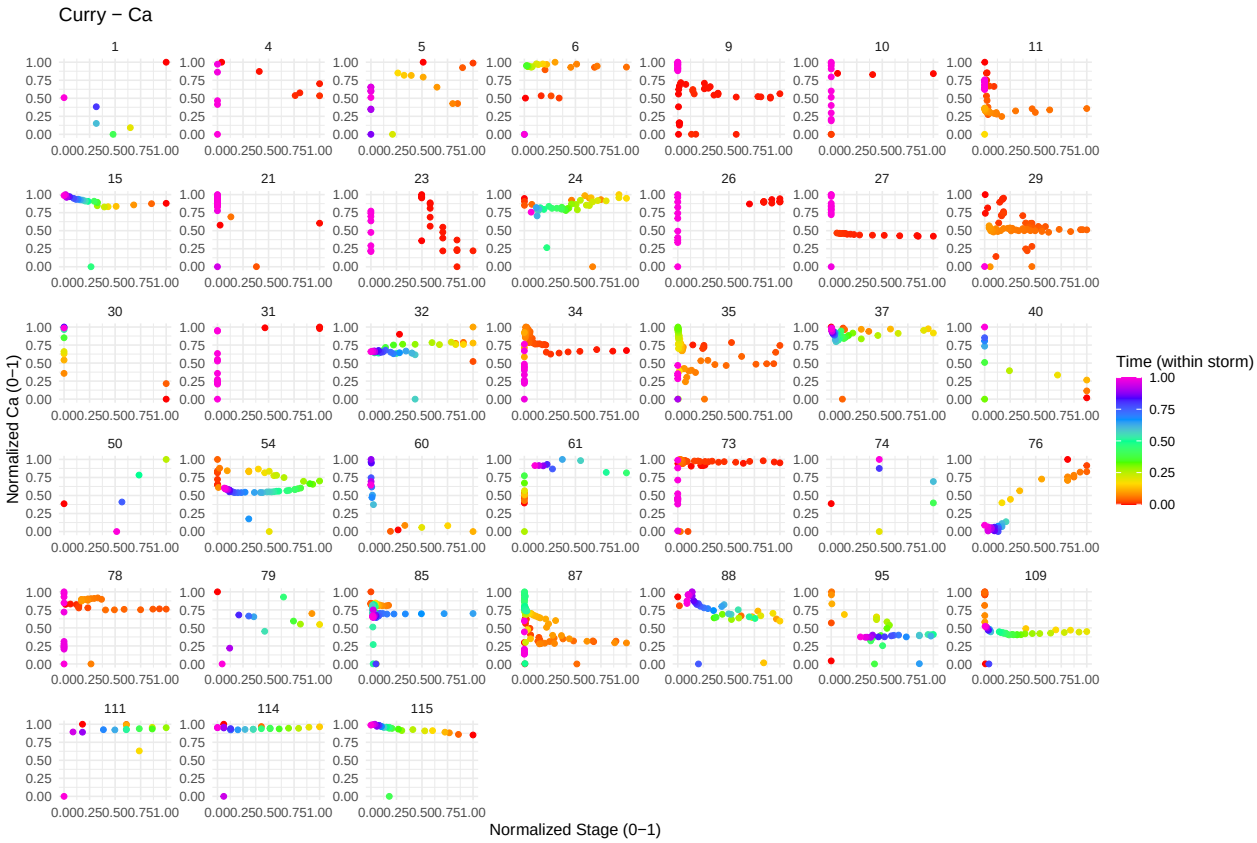


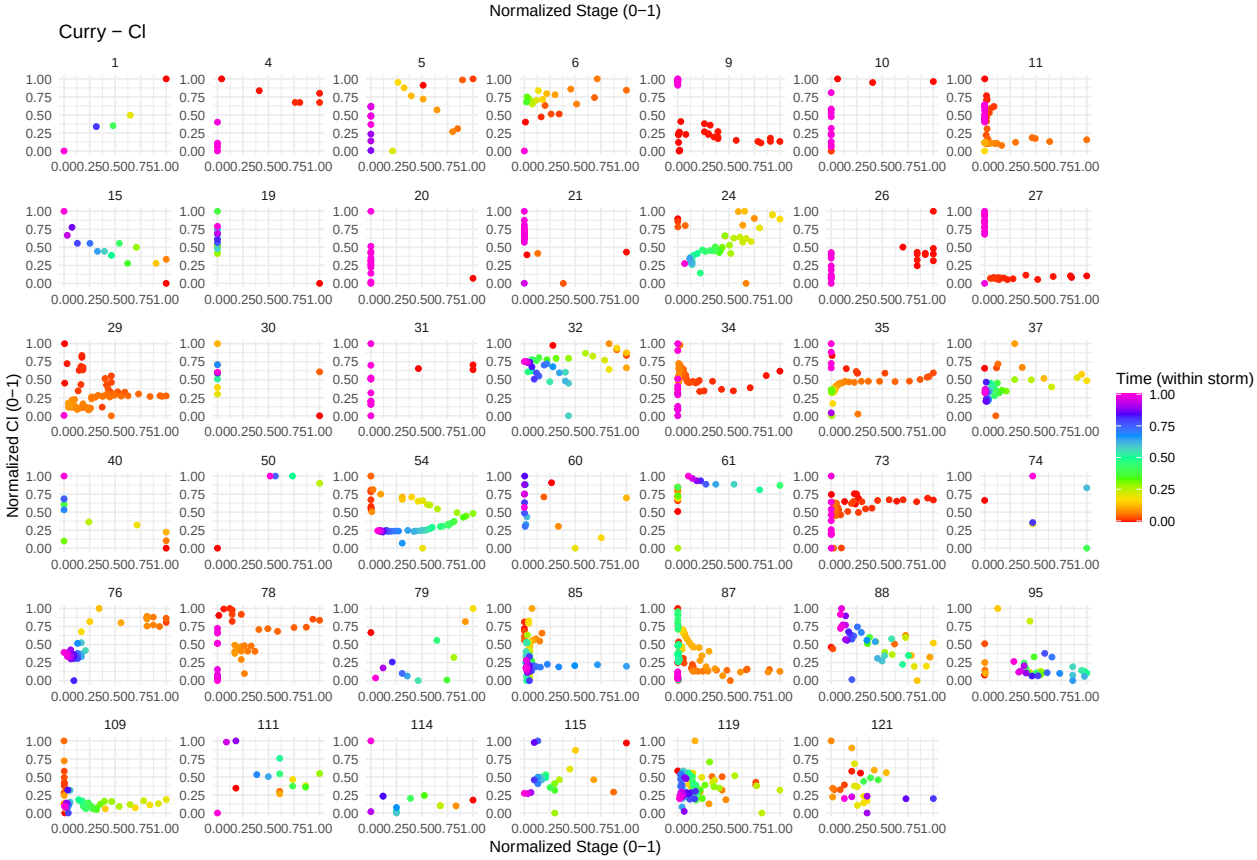
Lake Margarita − mean HI with confidence intervals per storm per analyte

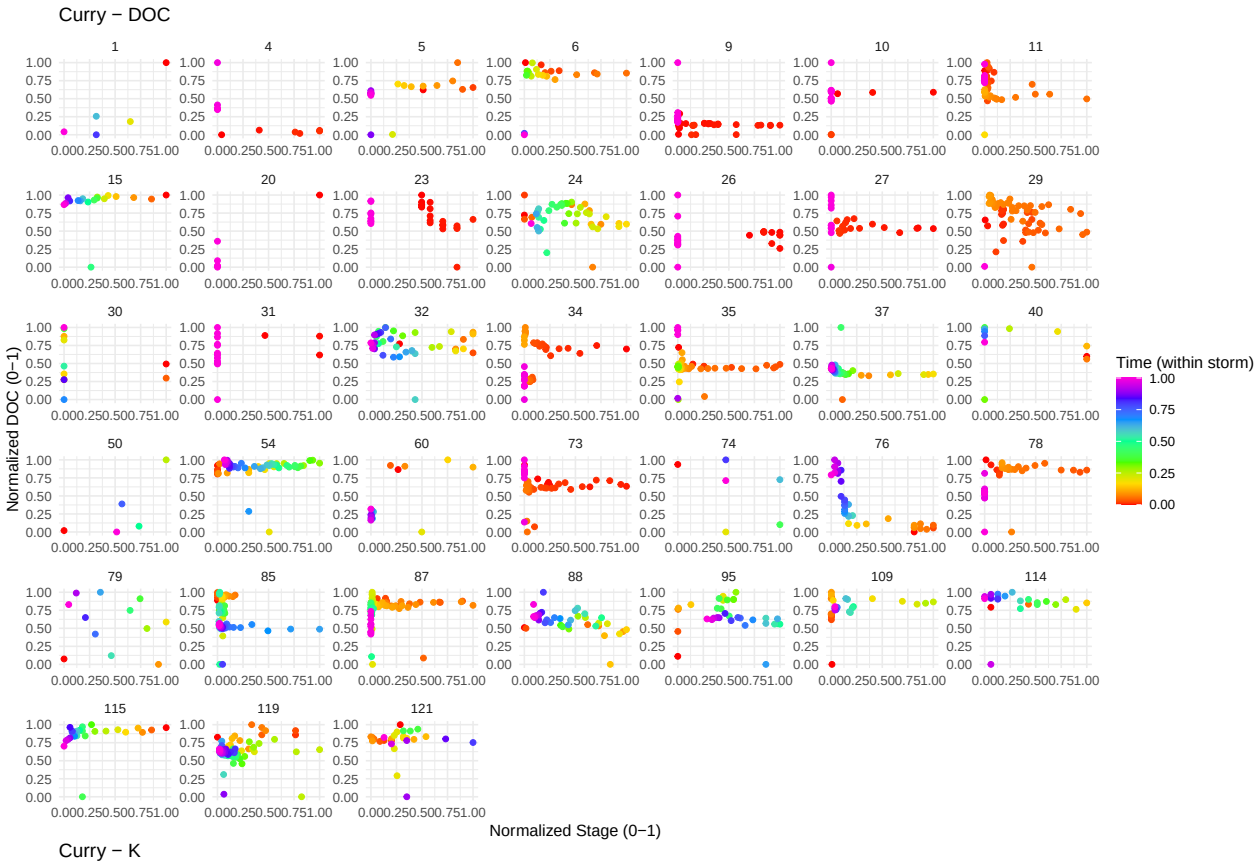


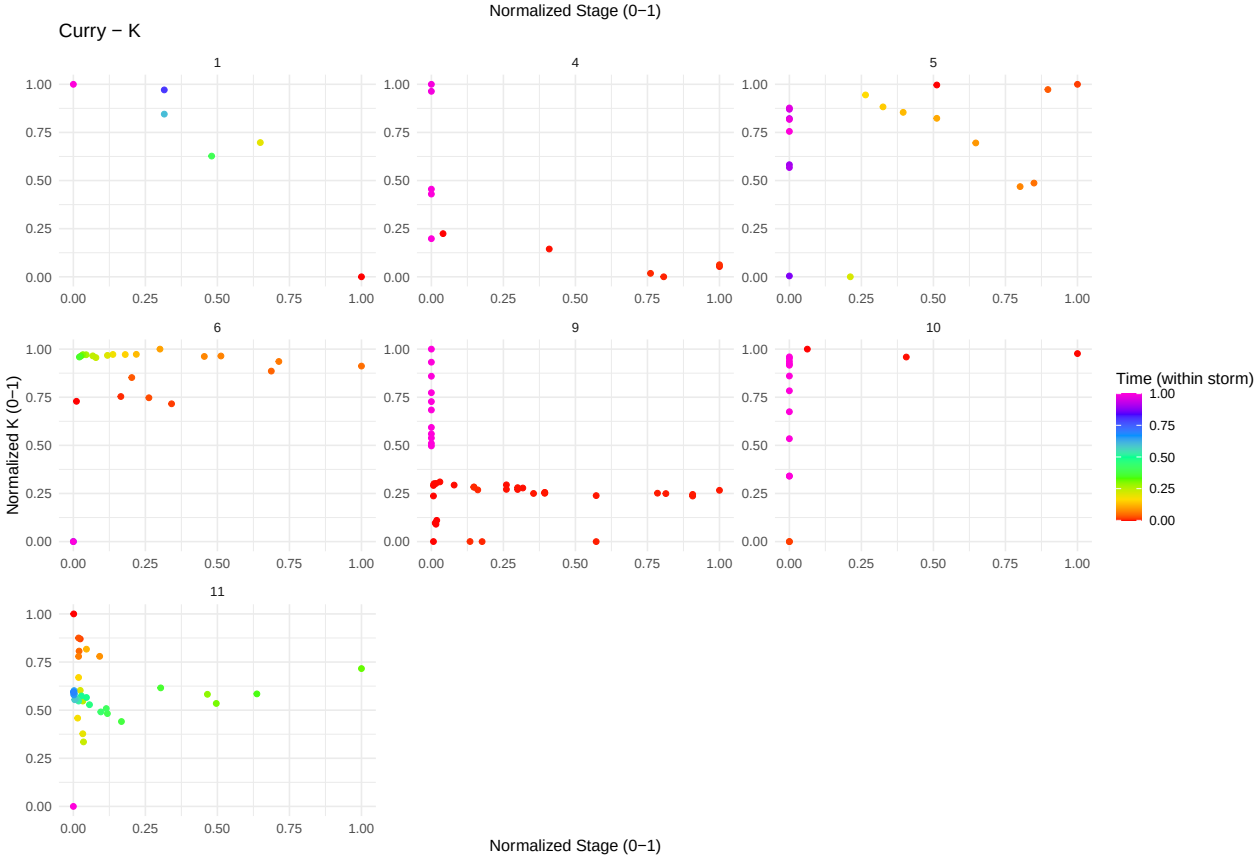
Plotting Beta

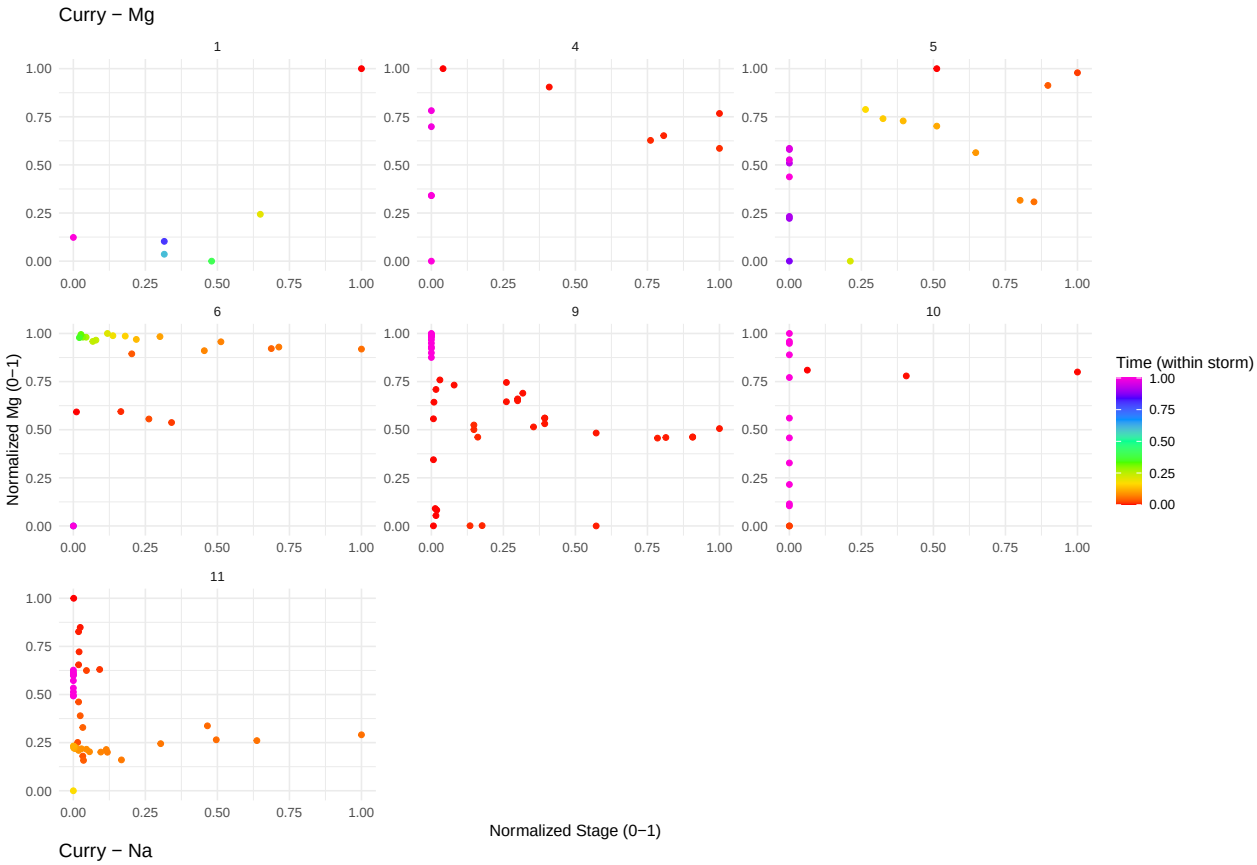
Plotting CQ

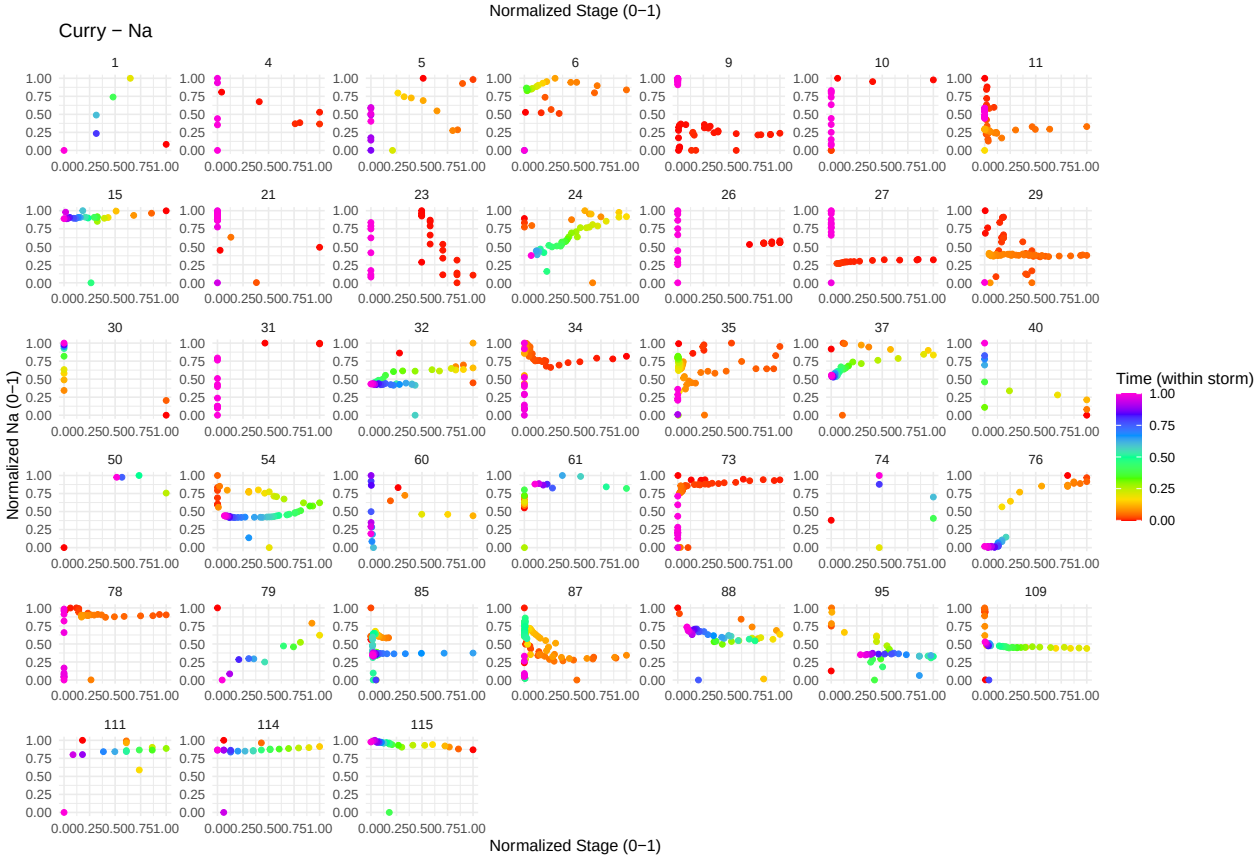

Curry − Ca
Normalized Ca (0−1)
Normalized Stage (0−1)
Time (within storm)

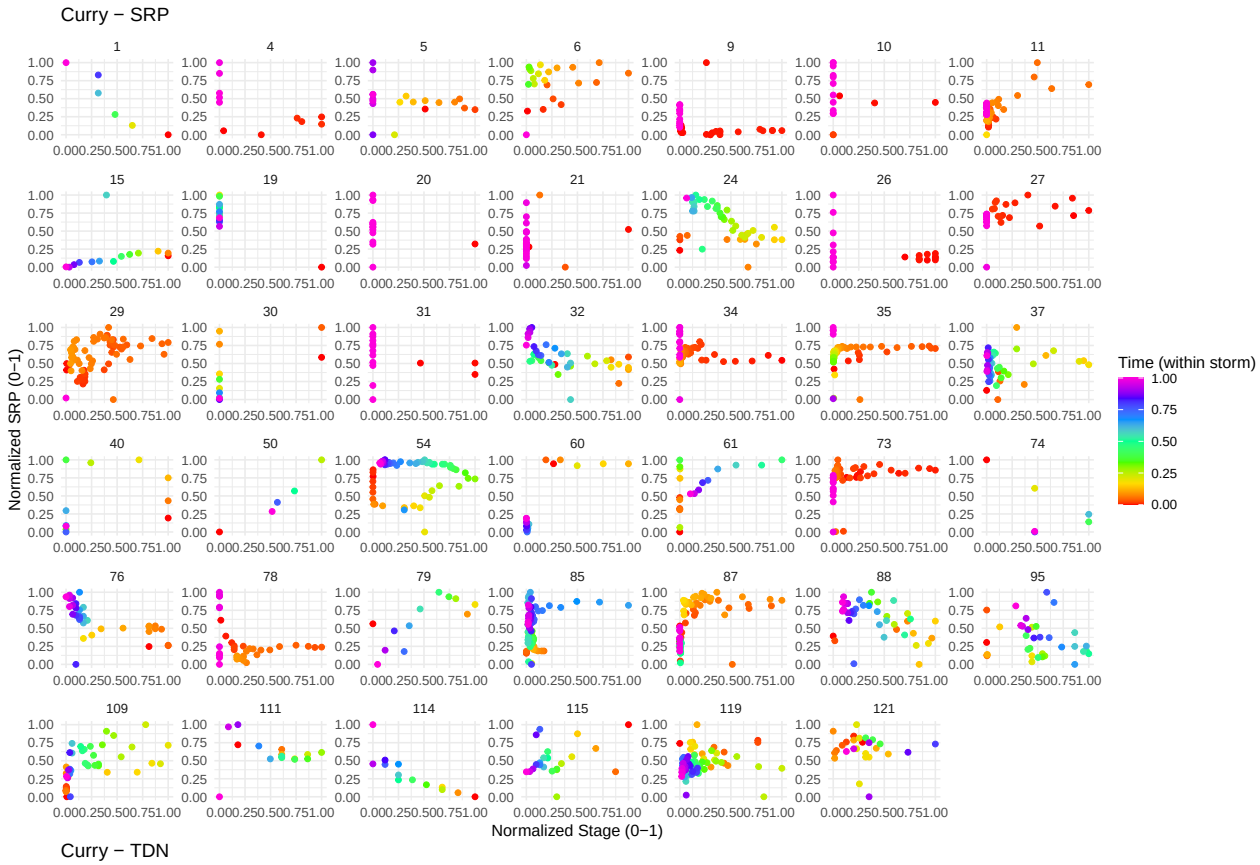

Curry − Cl
Normalized Cl (0−1)
Normalized Stage (0−1)
Time (within storm)

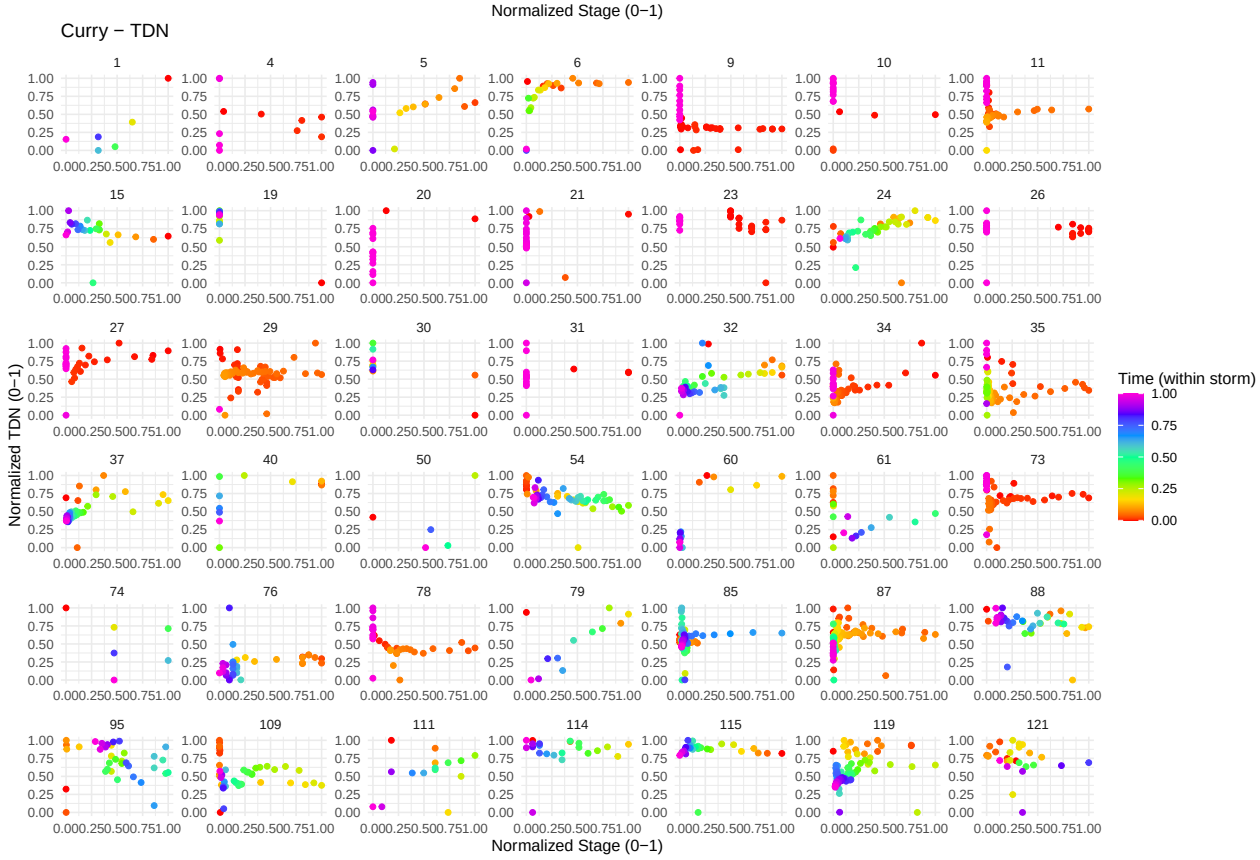

Curry − DOC
Normalized DOC (0−1)
Normalized Stage (0−1)
Time (within storm)

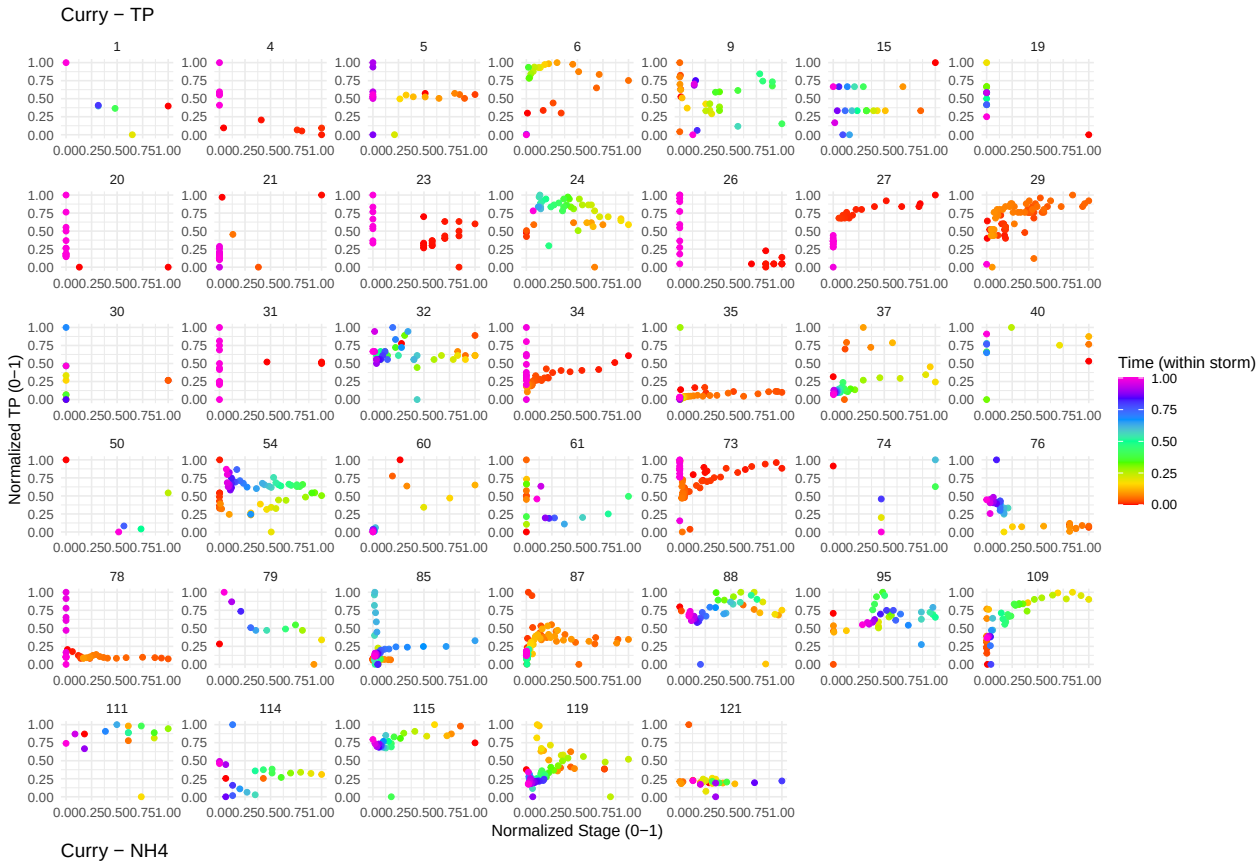

Curry − K
Normalized K (0−1)
Normalized Stage (0−1)
Time (within storm)

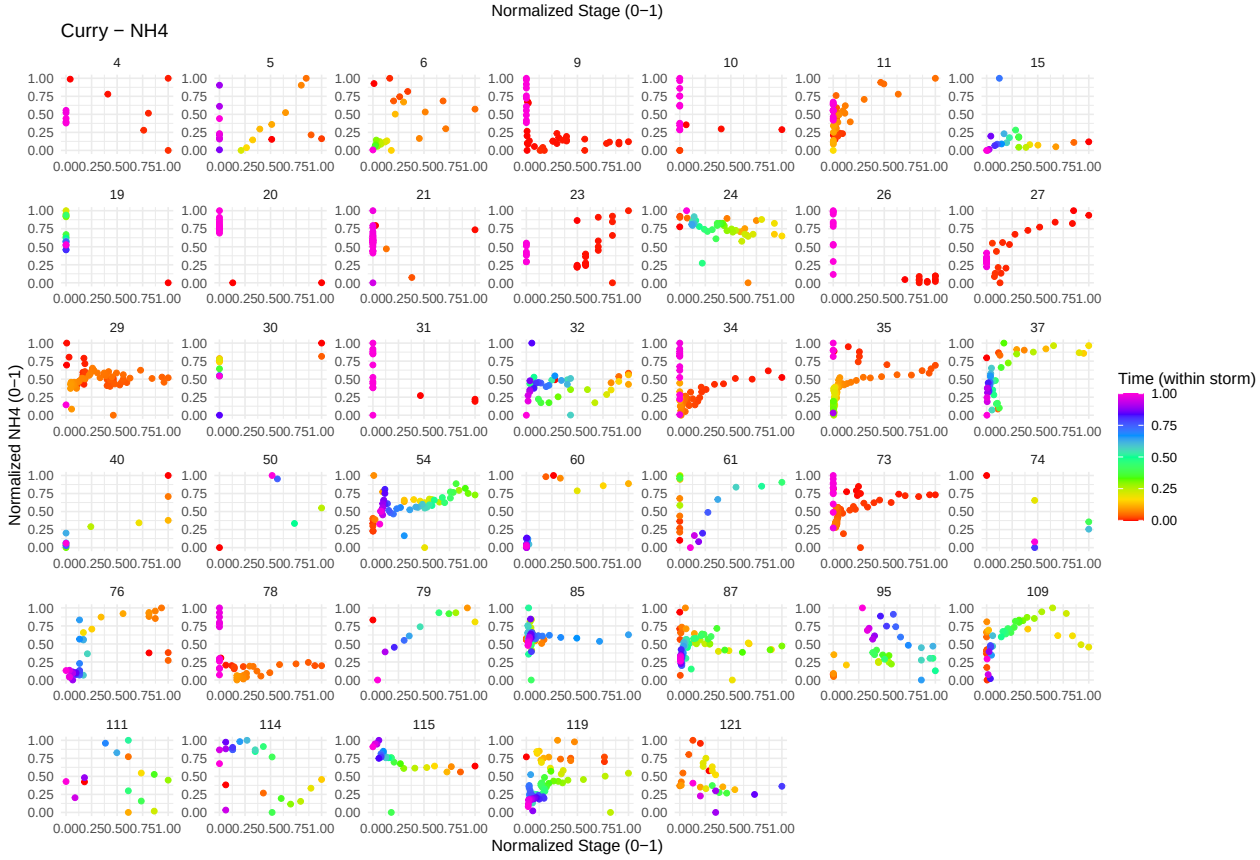

Curry − Mg
Normalized Mg (0−1)
Normalized Stage (0−1)
Time (within storm)

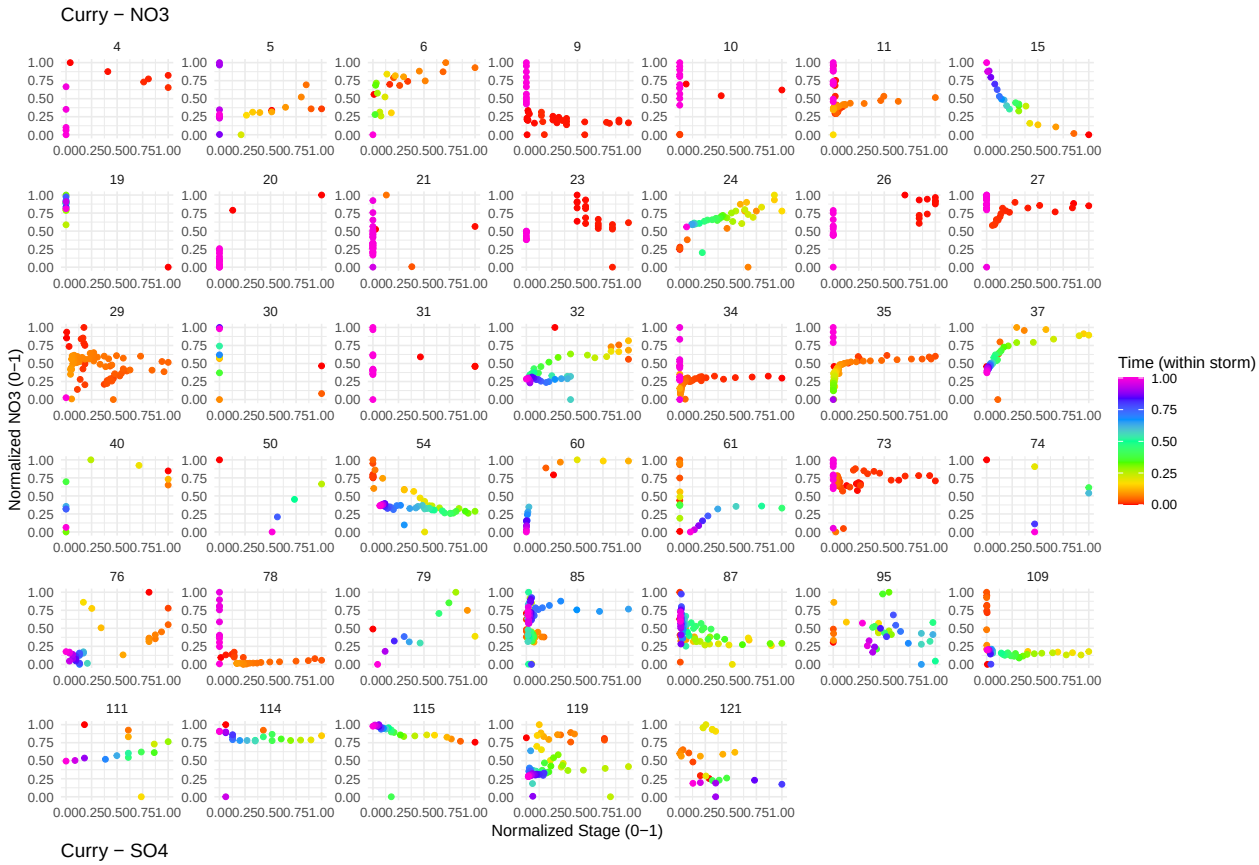

Curry − Na
Normalized Na (0−1)
Normalized Stage (0−1)
Time (within storm)

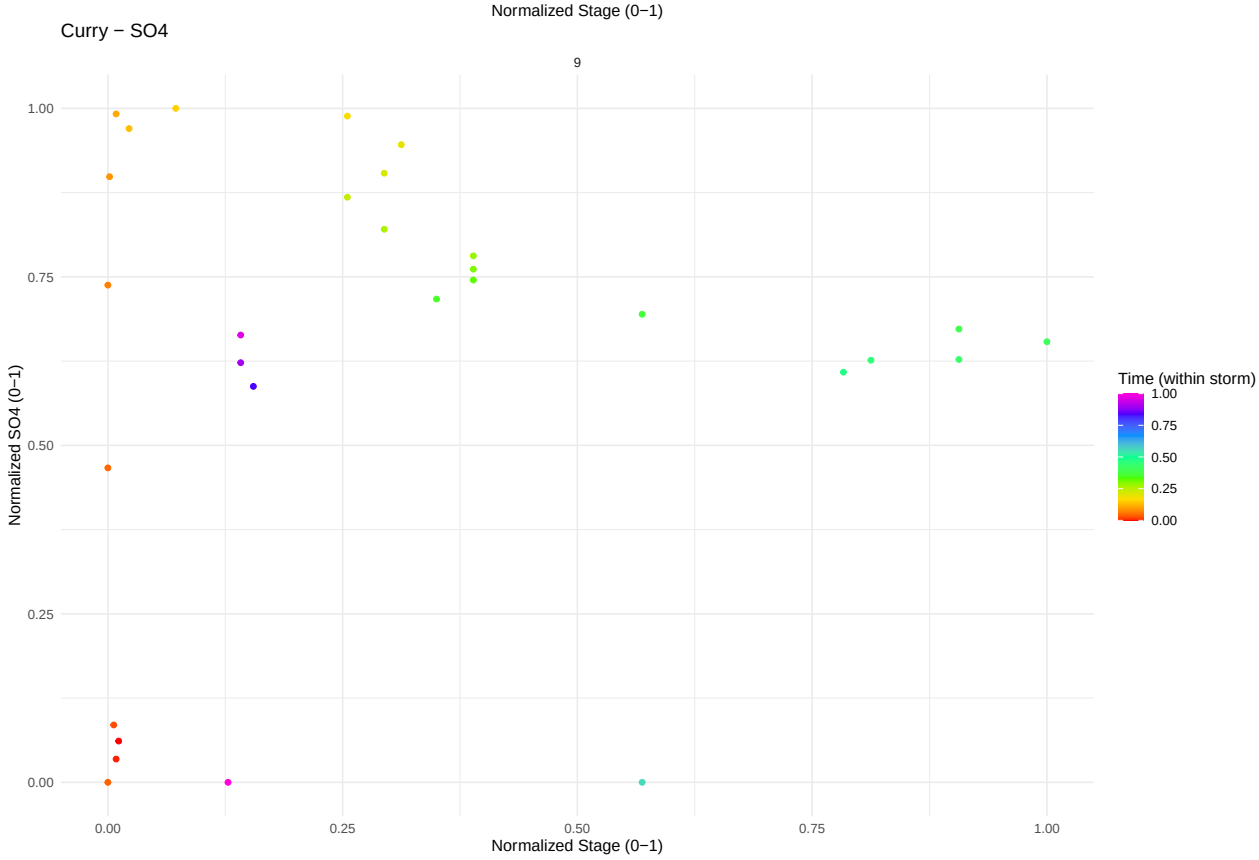

Curry − SRP
Normalized SRP (0−1)
Normalized Stage (0−1)
Time (within storm)

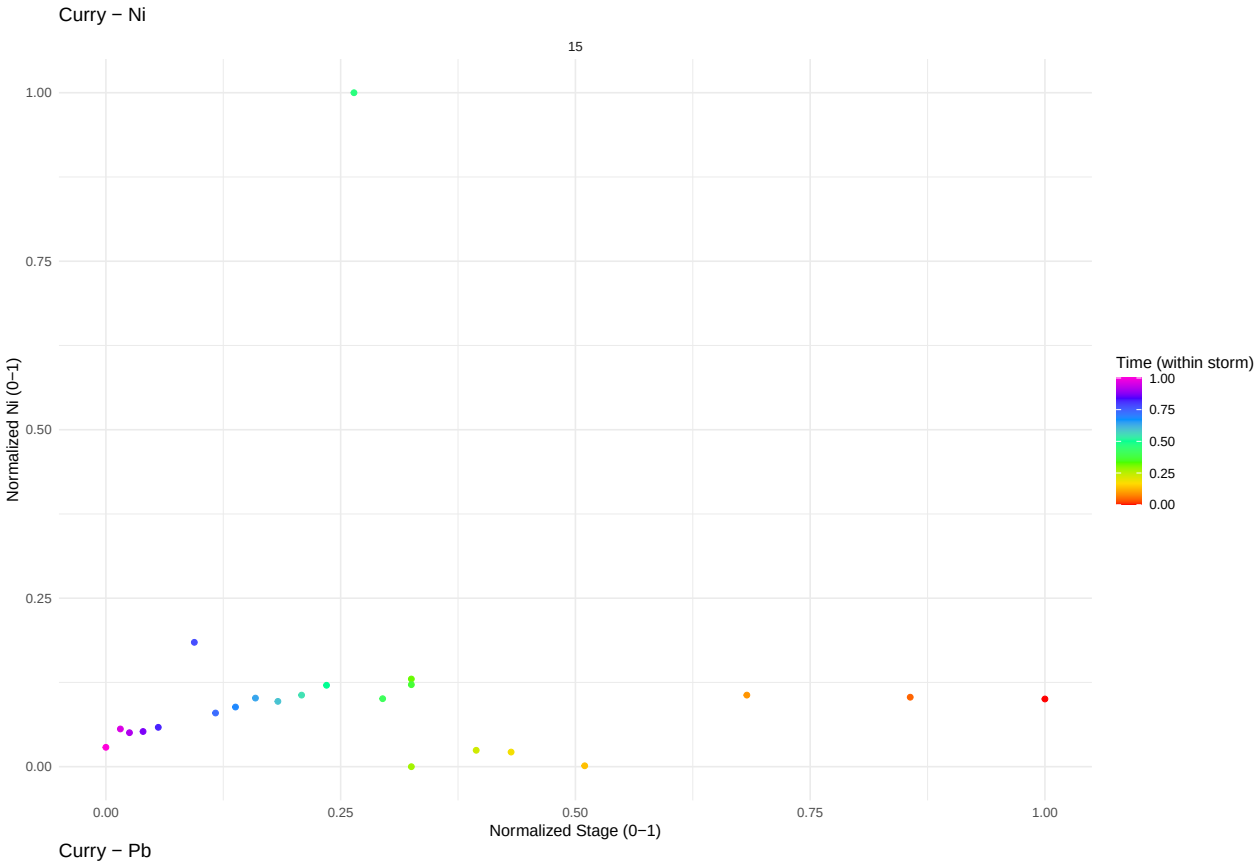

Curry − TDN
Normalized TDN (0−1)
Normalized Stage (0−1)
Time (within storm)

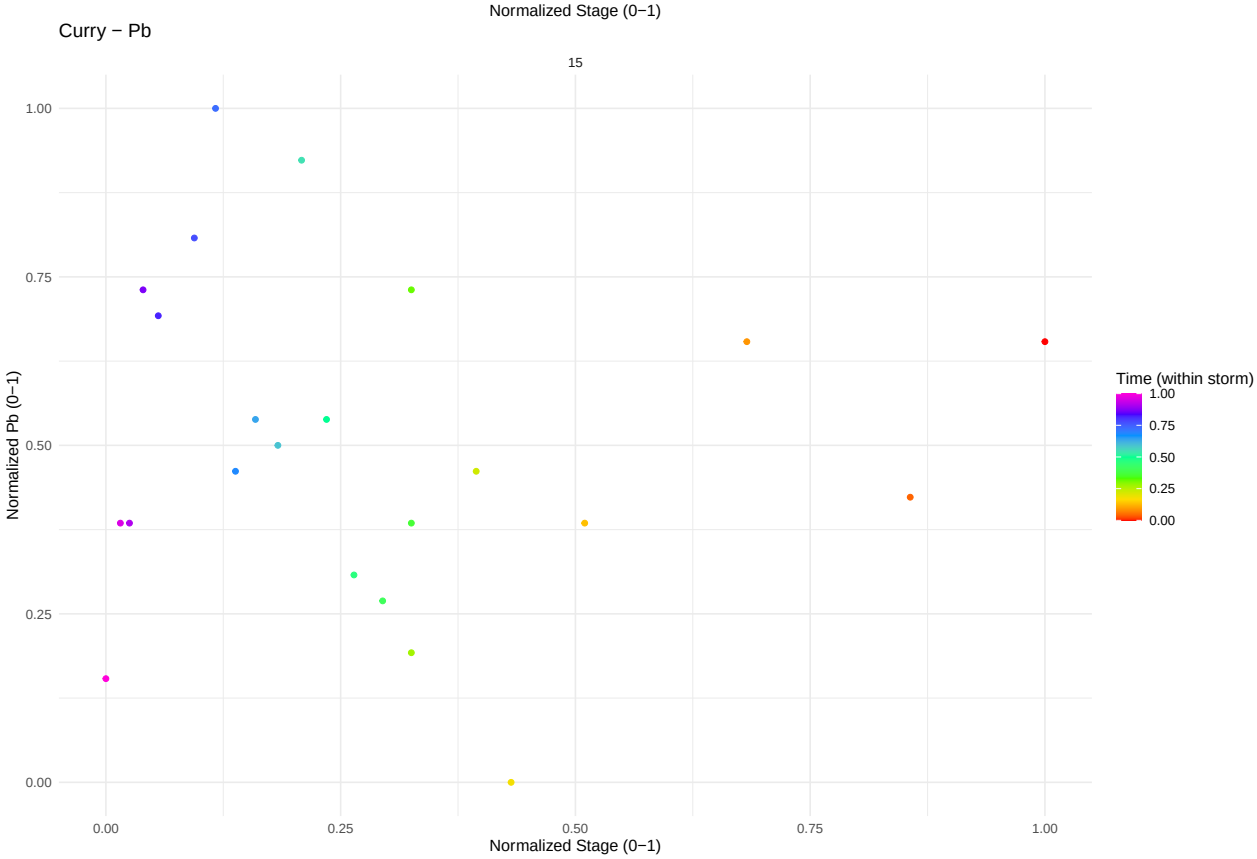

Curry − TP
Normalized TP (0−1)
Normalized Stage (0−1)
Time (within storm)

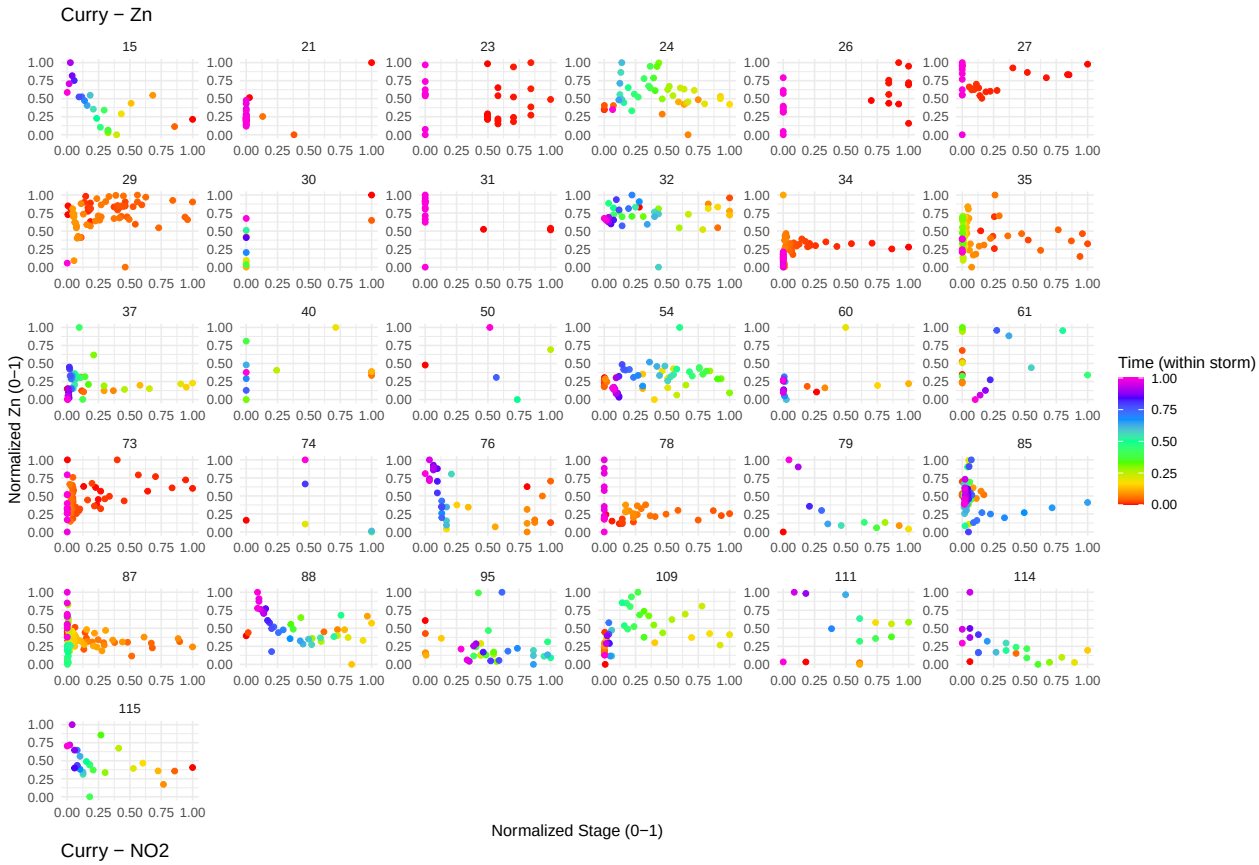

Curry − NH4
Normalized NH4 (0−1)
Normalized Stage (0−1)
Time (within storm)

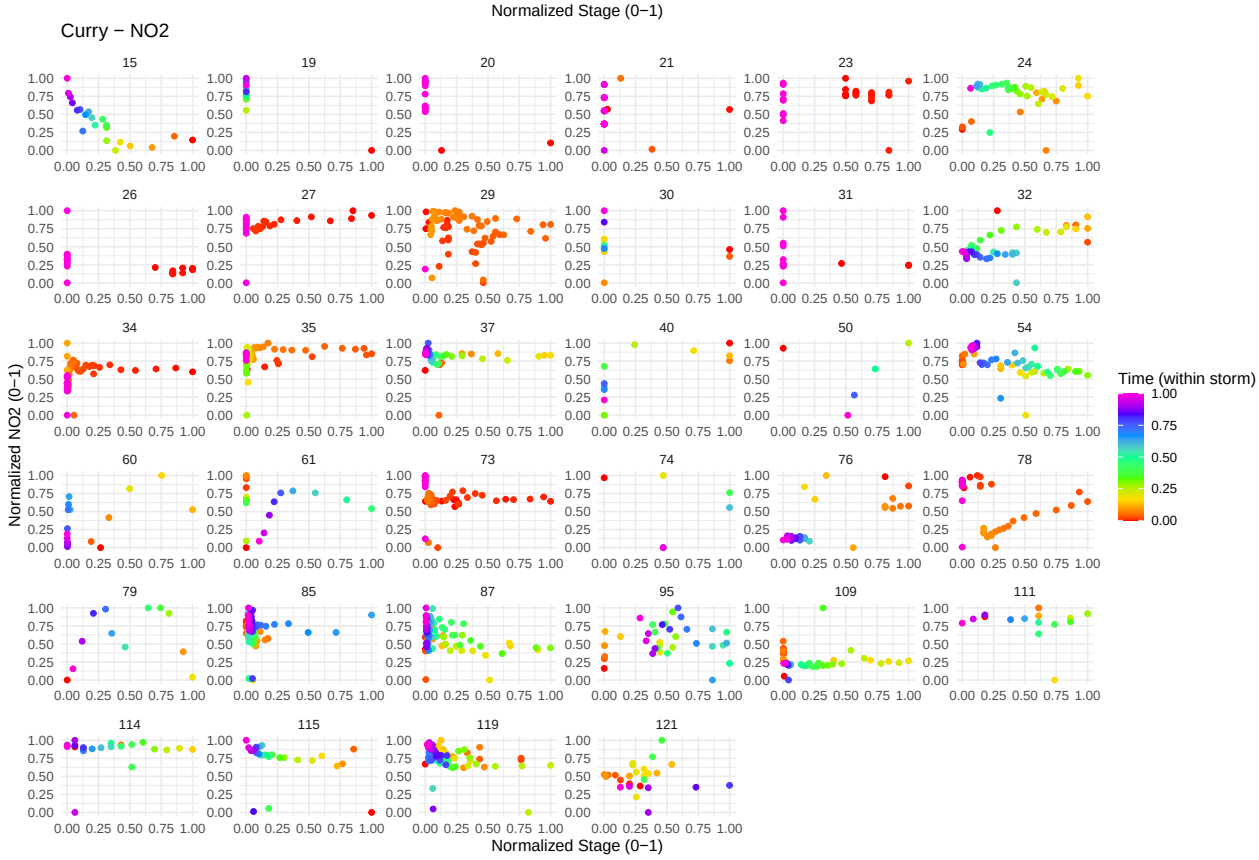

Curry − NO3
Normalized NO3 (0−1)
Normalized Stage (0−1)
Time (within storm)

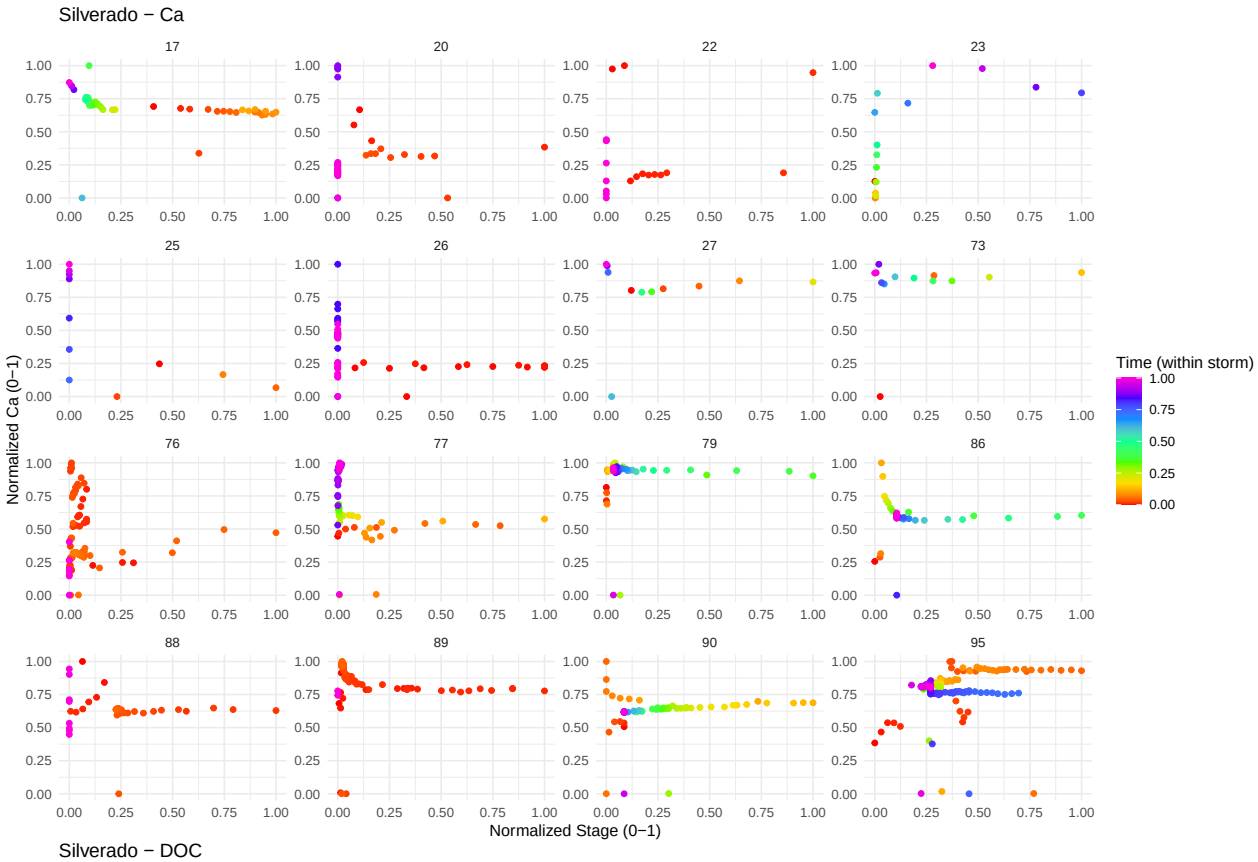

Curry − SO4
Normalized SO4 (0−1)
Normalized Stage (0−1)
Time (within storm)


Curry − Ni
15
Normalized Ni (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Curry − Pb
15
Normalized Pb (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

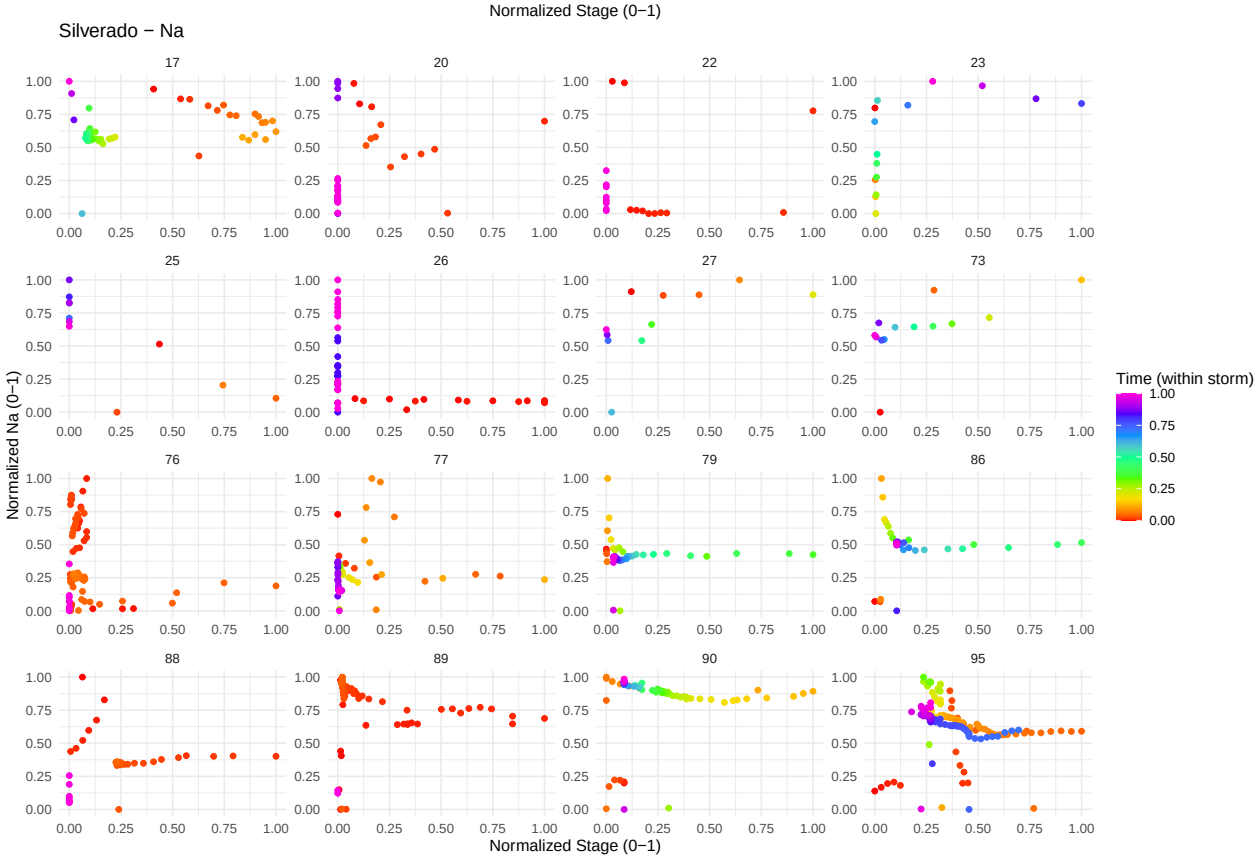

Curry − Zn
Normalized Zn (0−1)
Normalized Stage (0−1)
Time (within storm)


Curry − NO2
Normalized NO2 (0−1)
Normalized Stage (0−1)
Time (within storm)


Silverado − Ca
17
20
22
23
25
26
27
73
76
77
79
86
88
89
90
95
Normalized Ca (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

Silverado − DOC
17
20
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
Normalized DOC (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

Silverado − NO2
17
20
22
23
25
26
27
76
77
86
88
89
90
95
101
103
105
Normalized NO2 (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Silverado − Na
17
20
22
23
25
26
27
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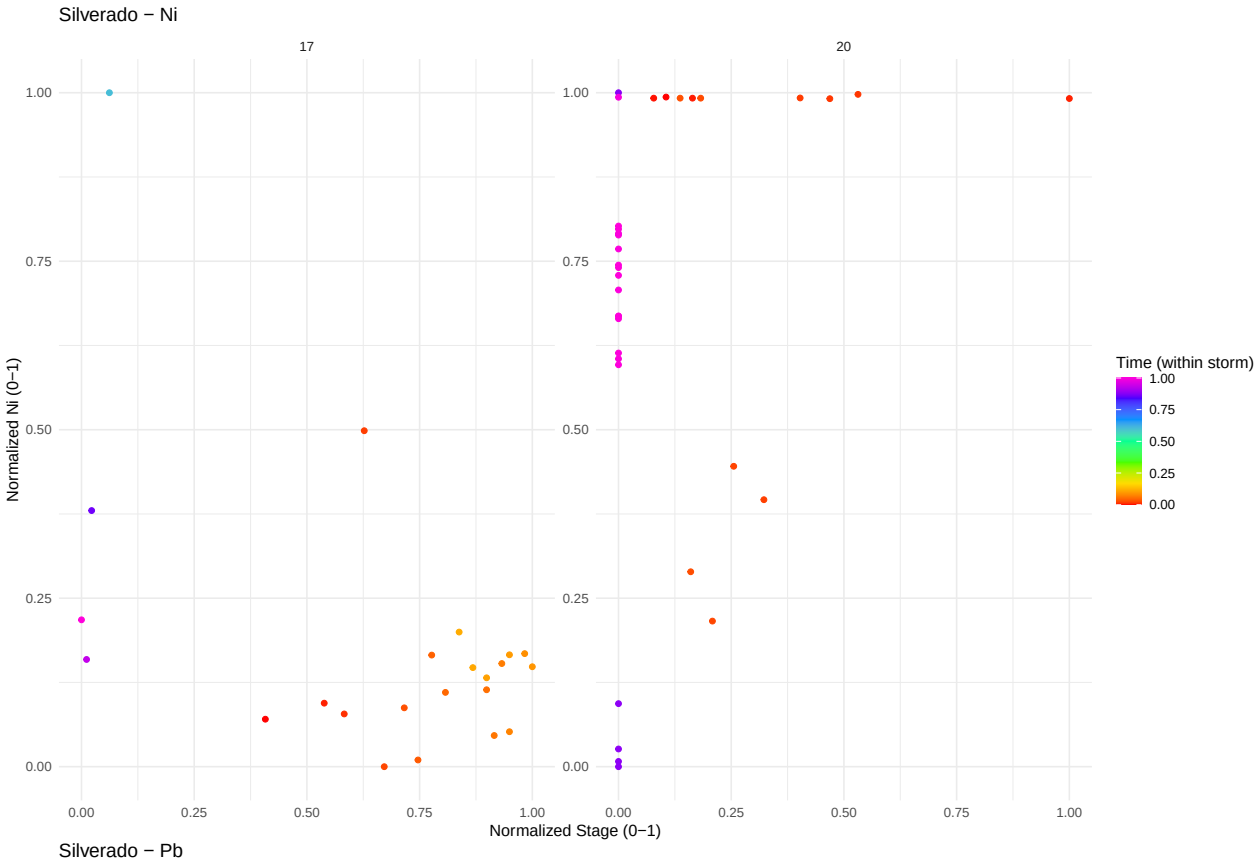
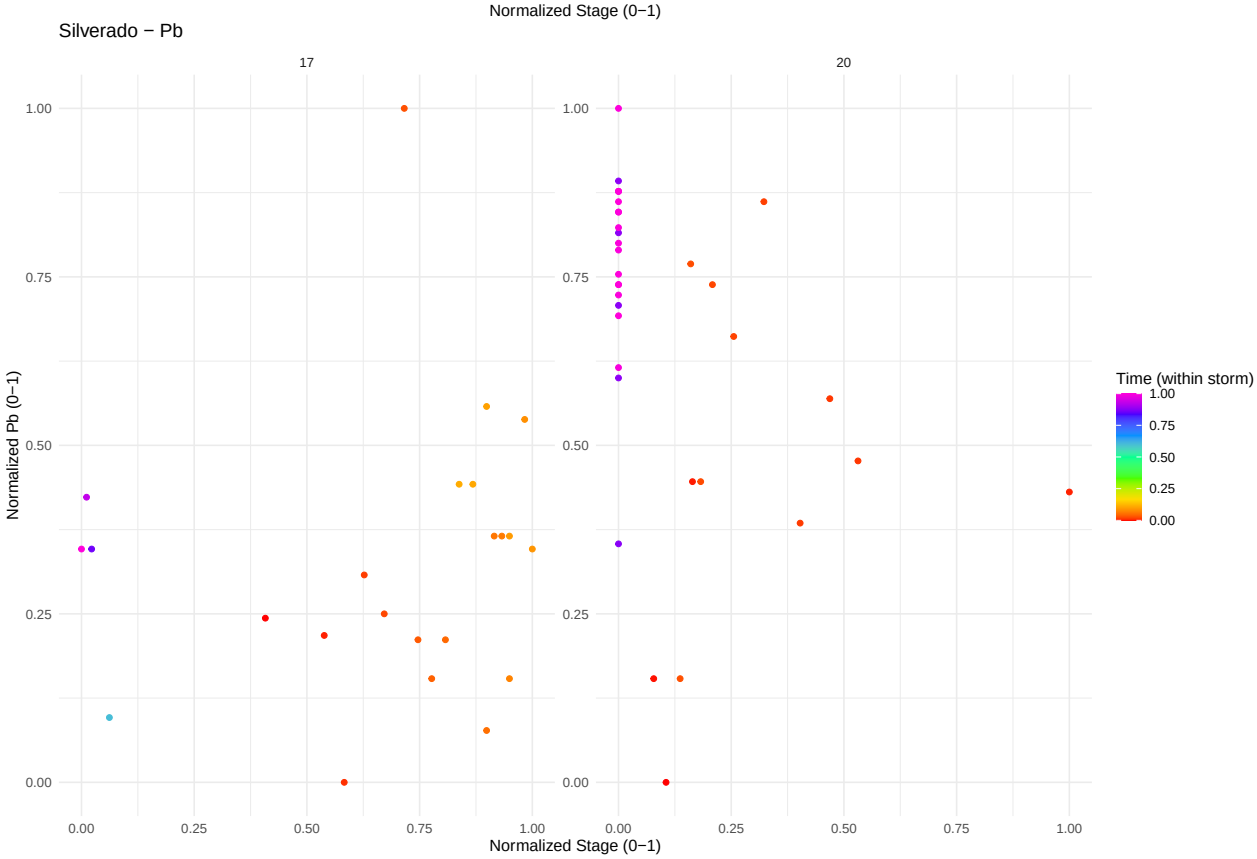
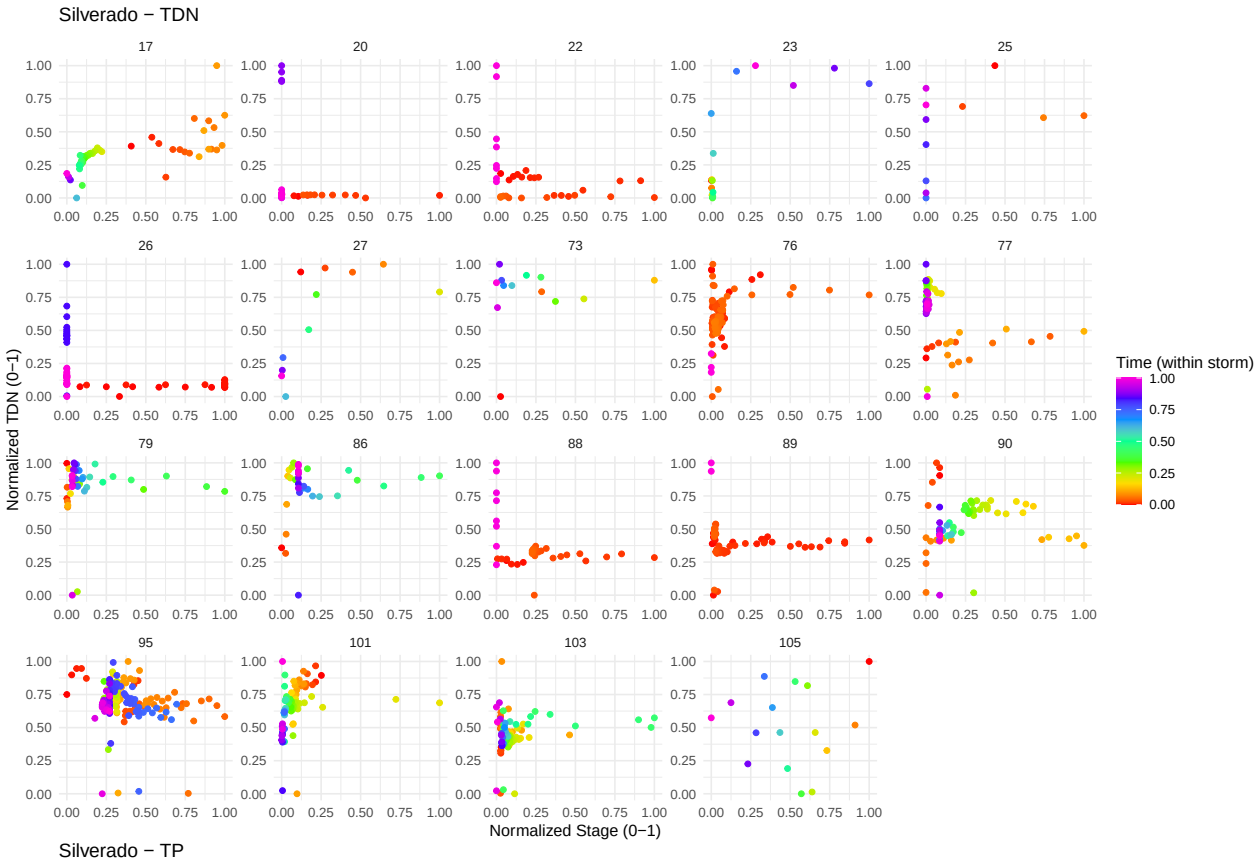
76
77
79
86
88
89
90
95
Normalized Na (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Silverado − Ni
17
20
Normalized Ni (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Silverado − Pb
17
20
Normalized Pb (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Silverado − TDN
17
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22
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27
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76
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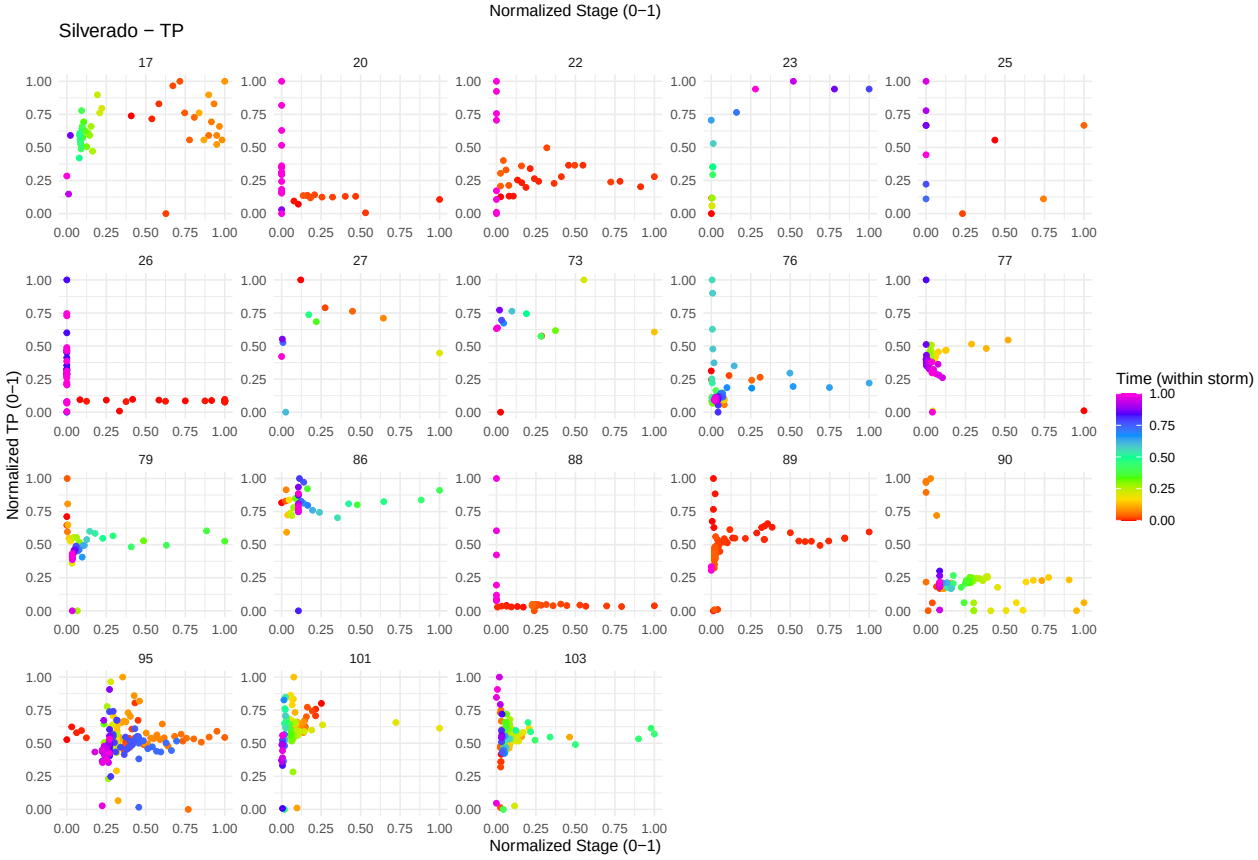
79
86
88
89
90
95
101
103
105
Normalized TDN (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Silverado − TP
17
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22
23
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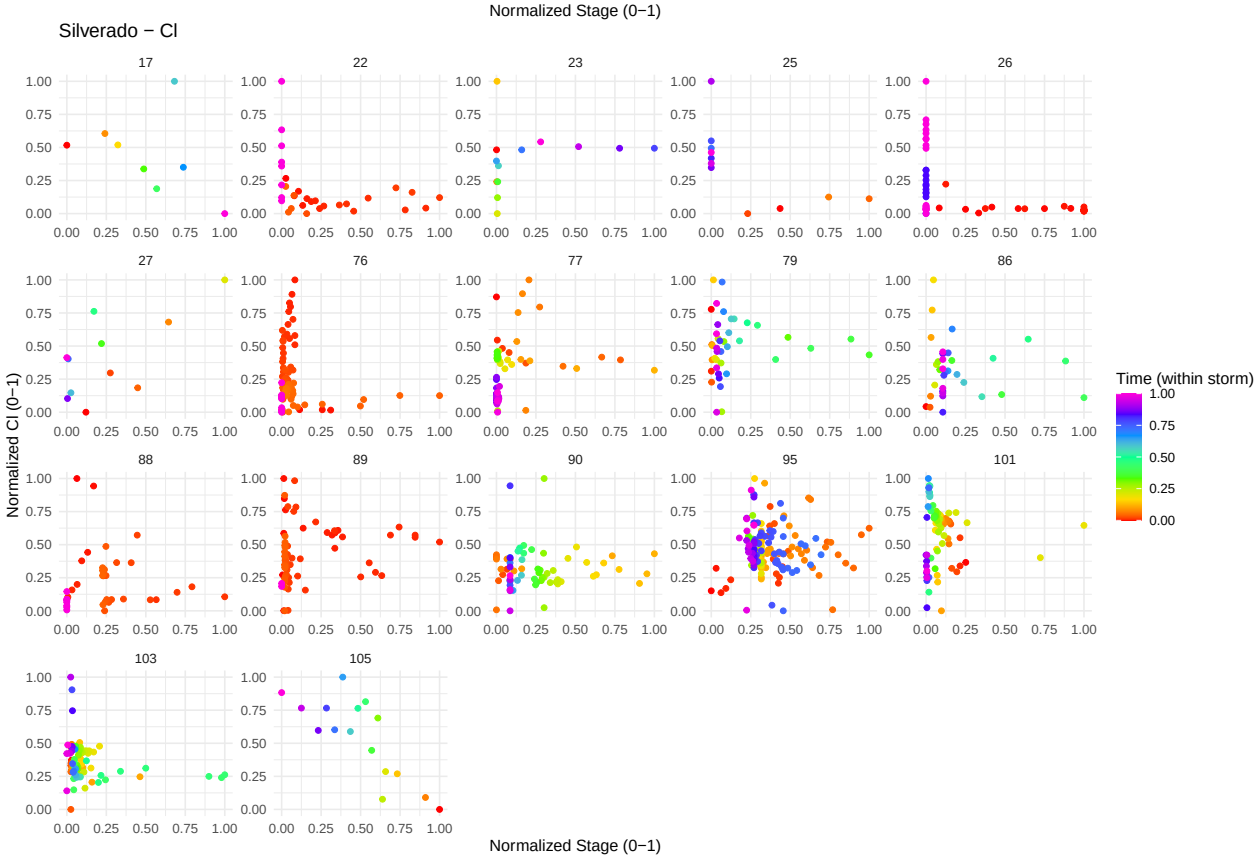
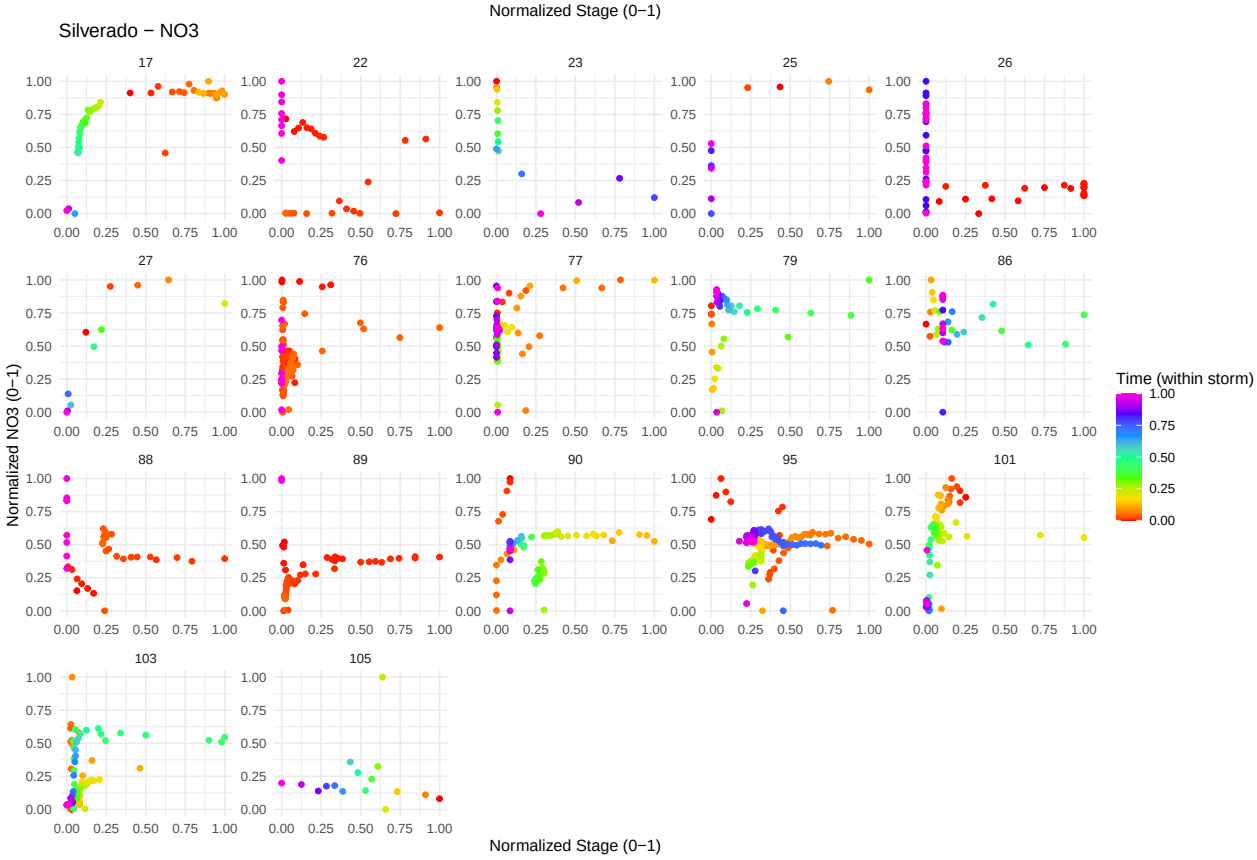
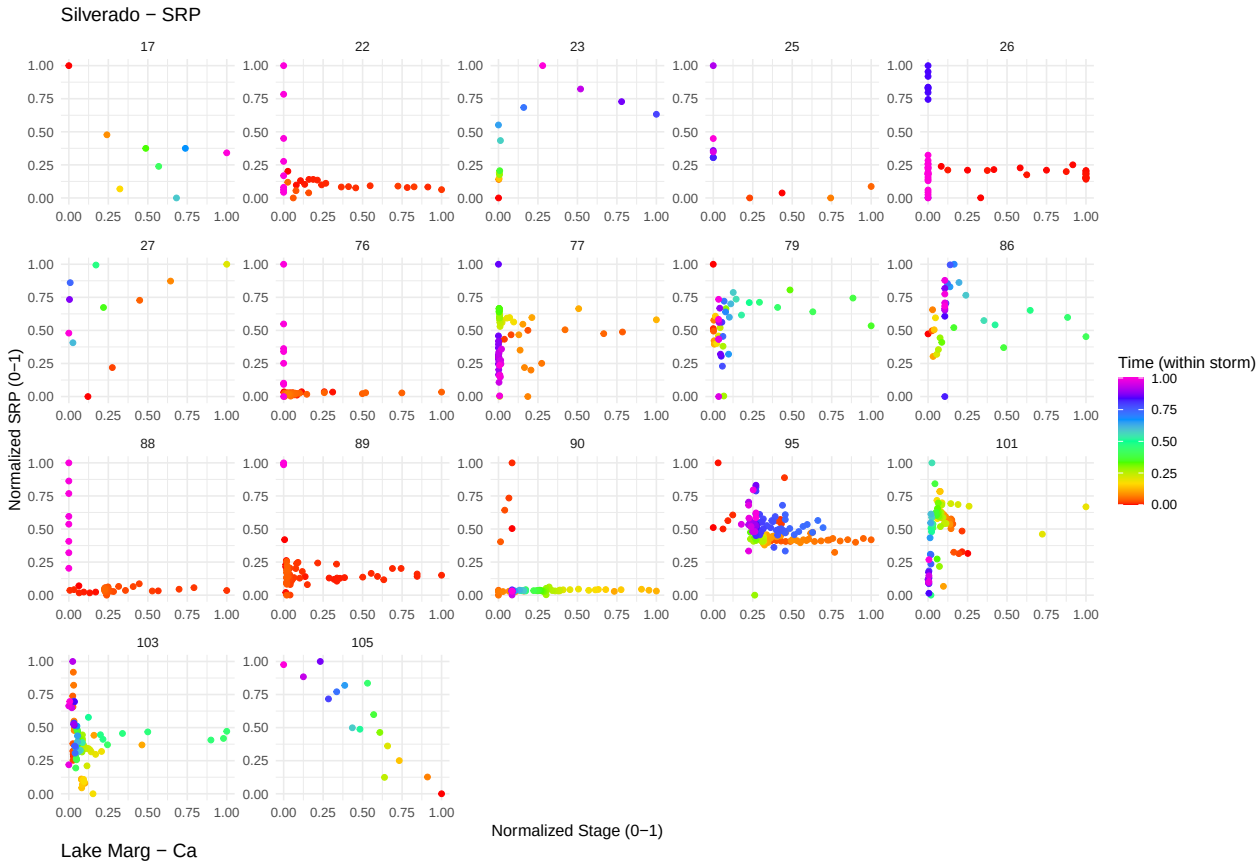
27
73
76
77
79
86
88
89
90
95
101
103
Normalized TP (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

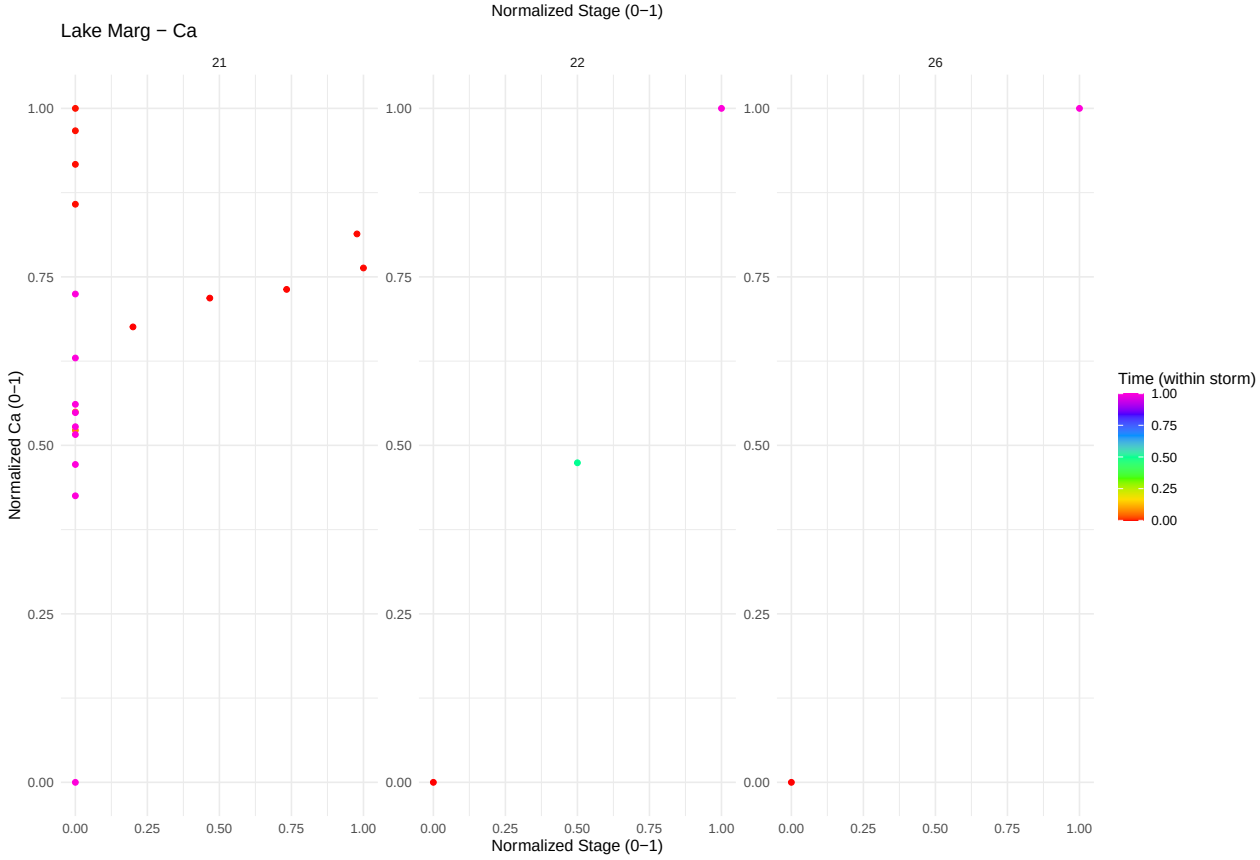
Silverado − Zn
17
20
22
23
25
26
27
73
76
77
79
86
88
89
90
95
Normalized Zn (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

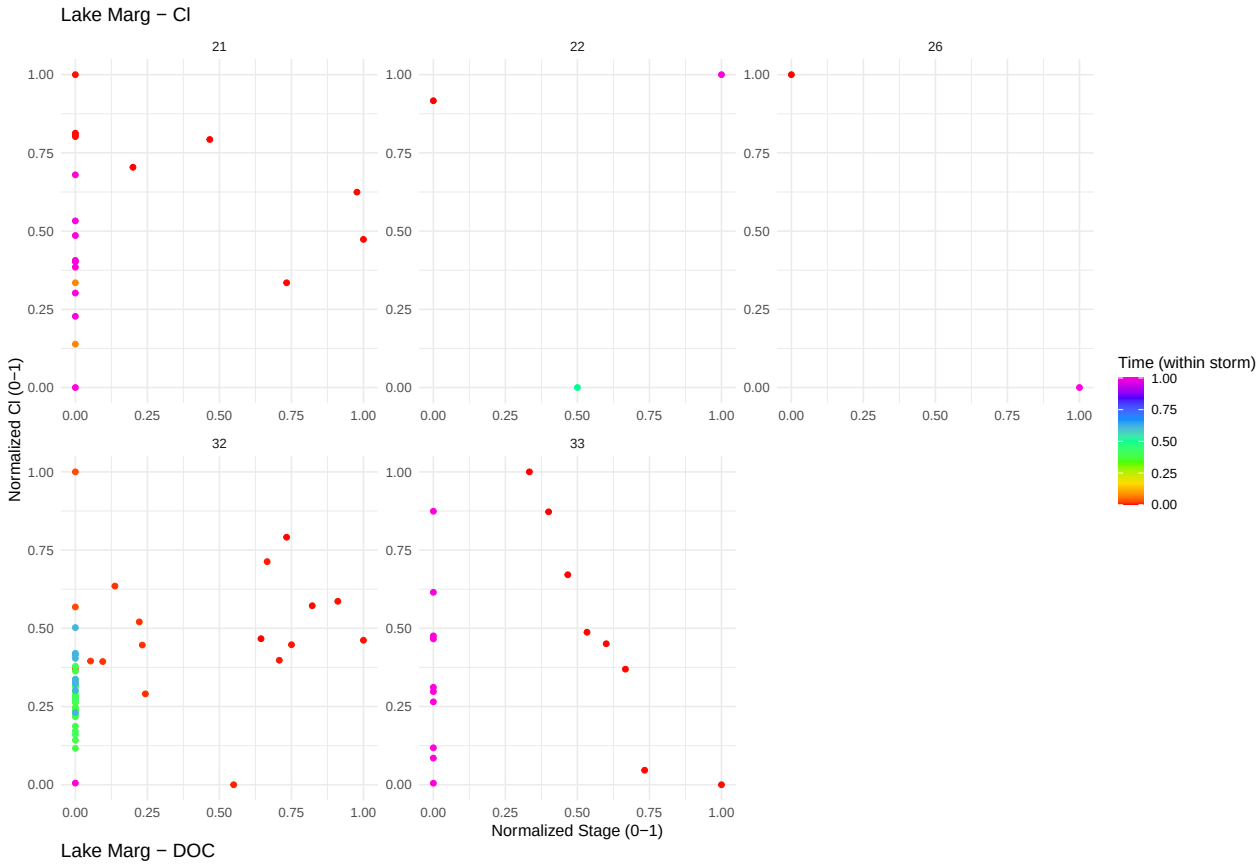

Silverado − Cl
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
Normalized Cl (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

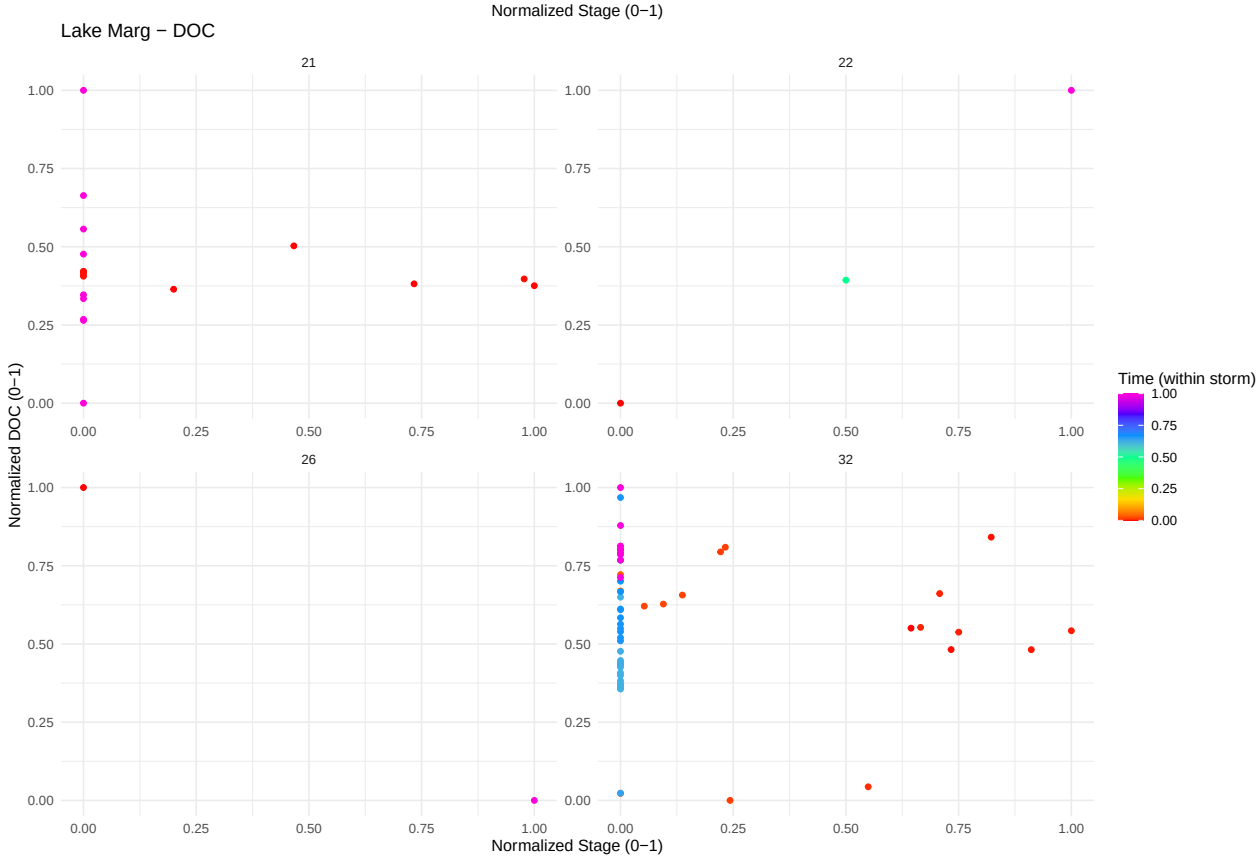
Silverado − NH4
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
Normalized NH4 (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

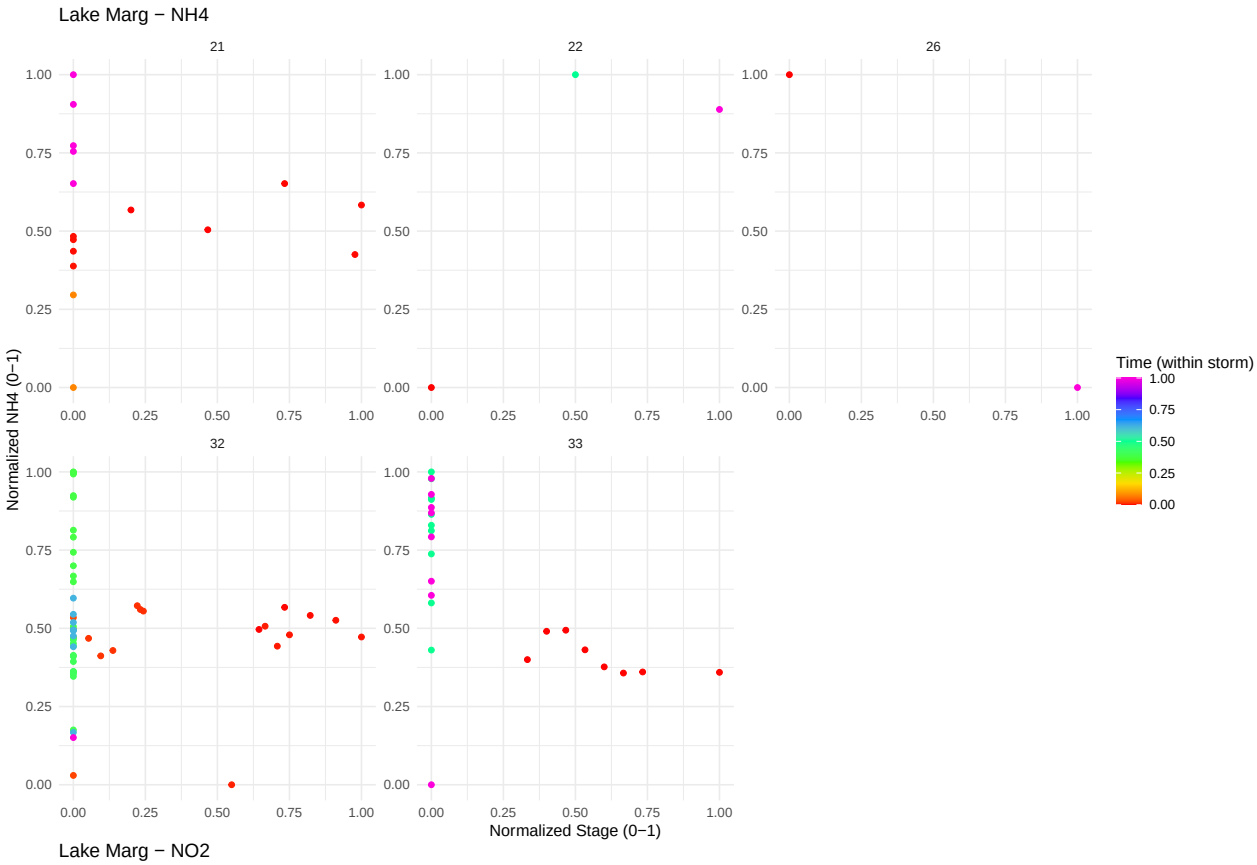

Silverado − NO3
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
Normalized NO3 (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

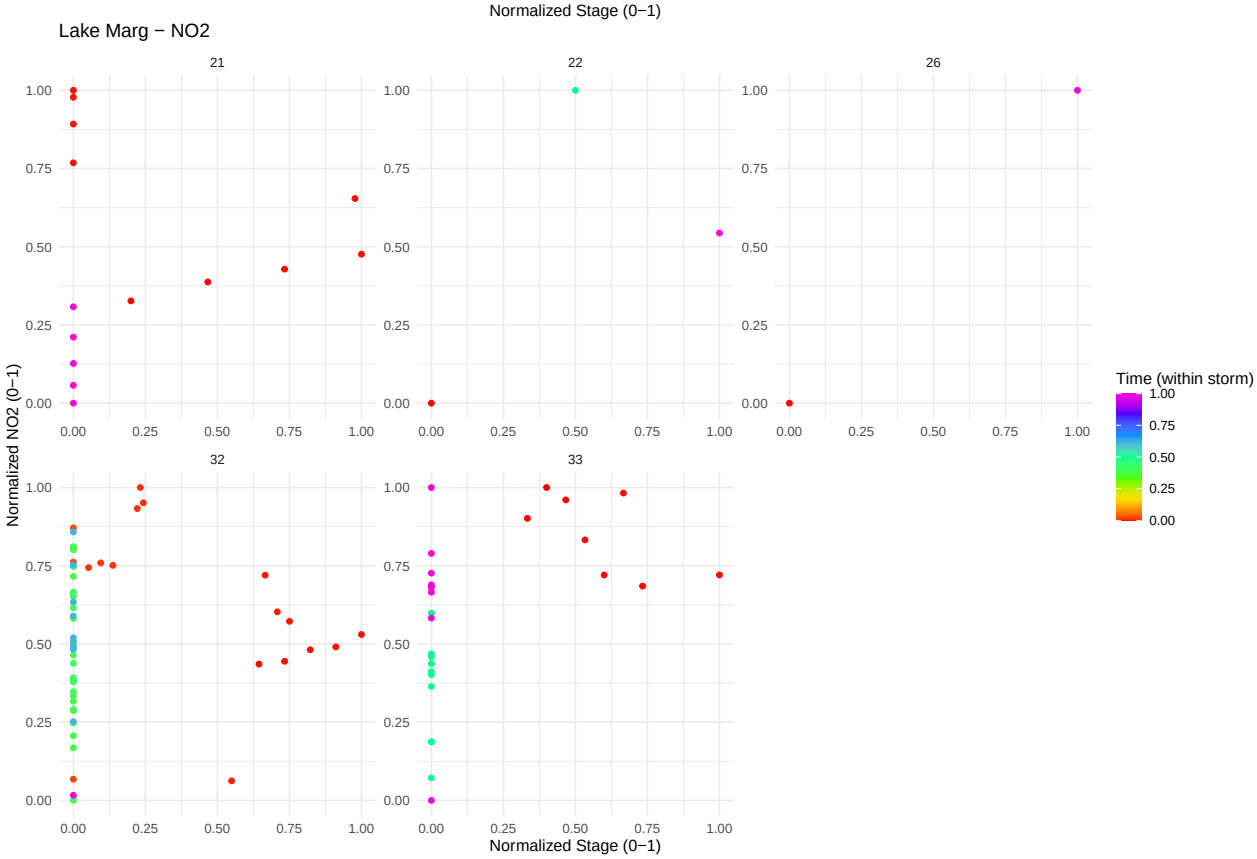

Silverado − SRP
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
Normalized SRP (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

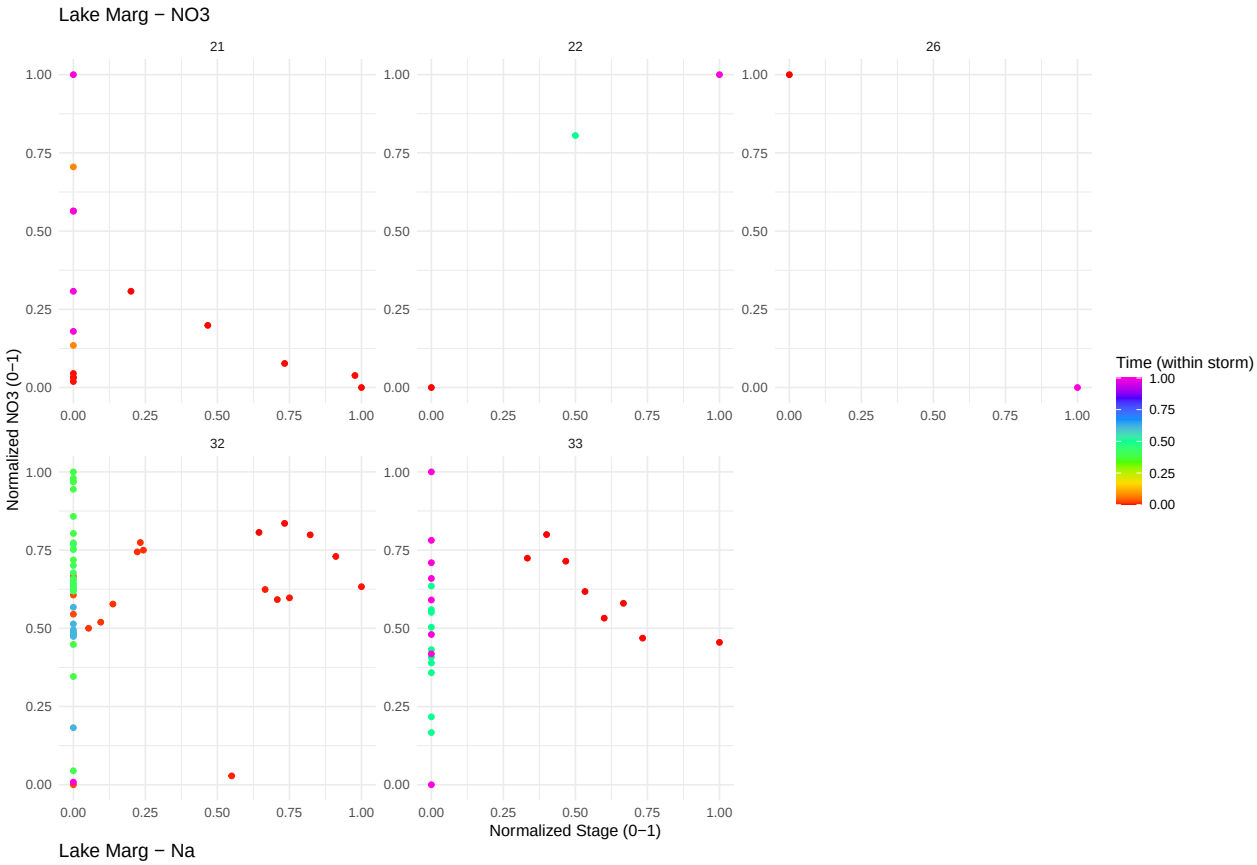

Lake Marg − Ca
21
22
26
Normalized Ca (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00

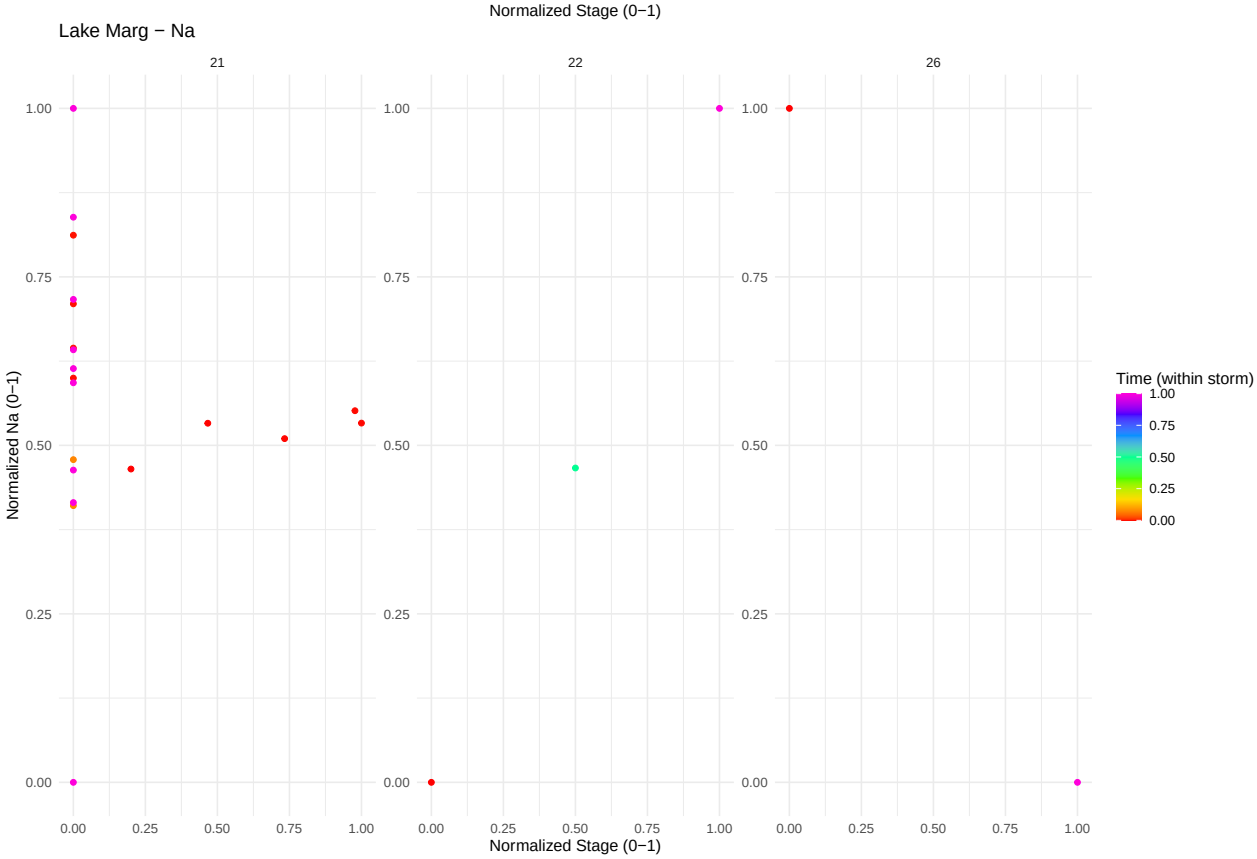

Lake Marg − Cl
21
22
26
32
33
Normalized Cl (0−1)
Normalized Stage (0−1)
Time (within storm)
0.00
0.25
0.50
0.75
1.00

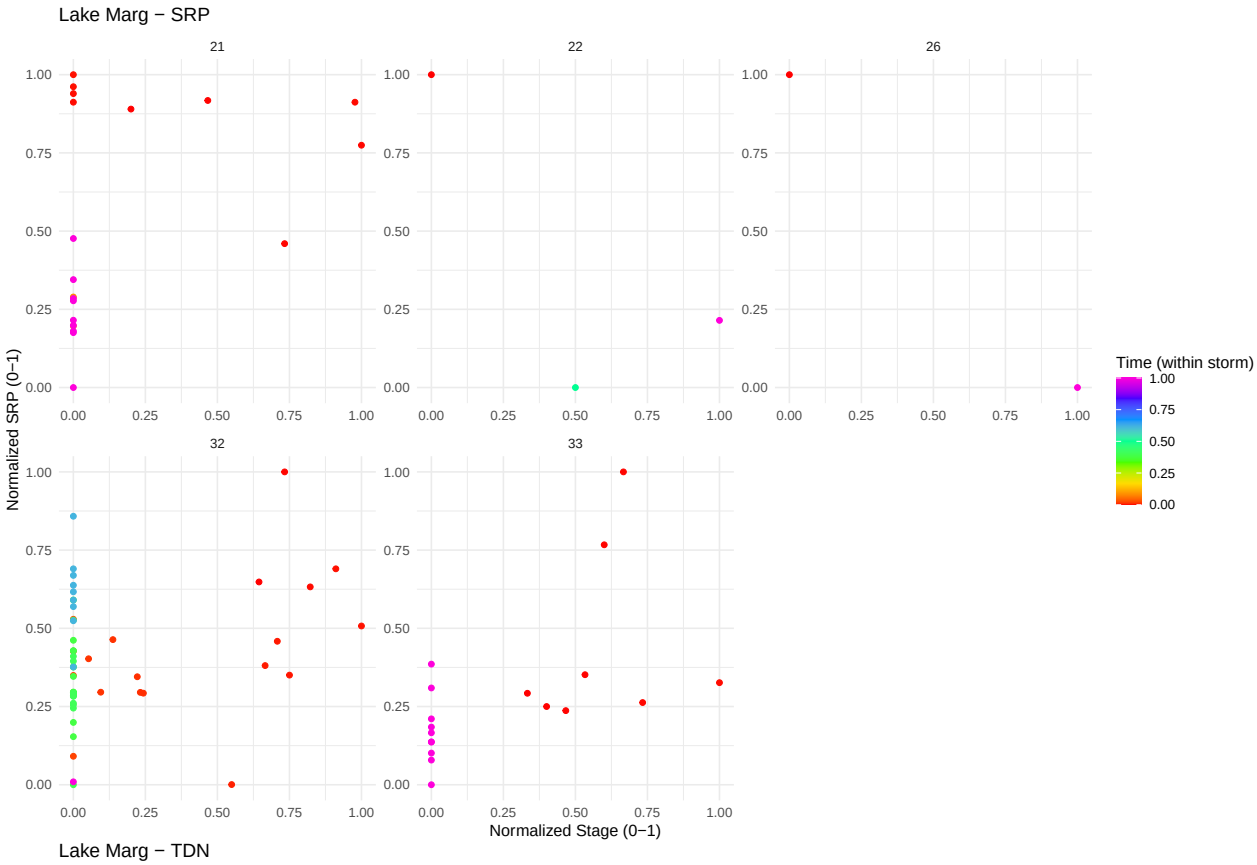

Lake Marg − DOC
21
22
26
32
Normalized DOC (0−1)
Normalized Stage (0−1)
Time (within storm)
0.00
0.25
0.50
0.75
1.00

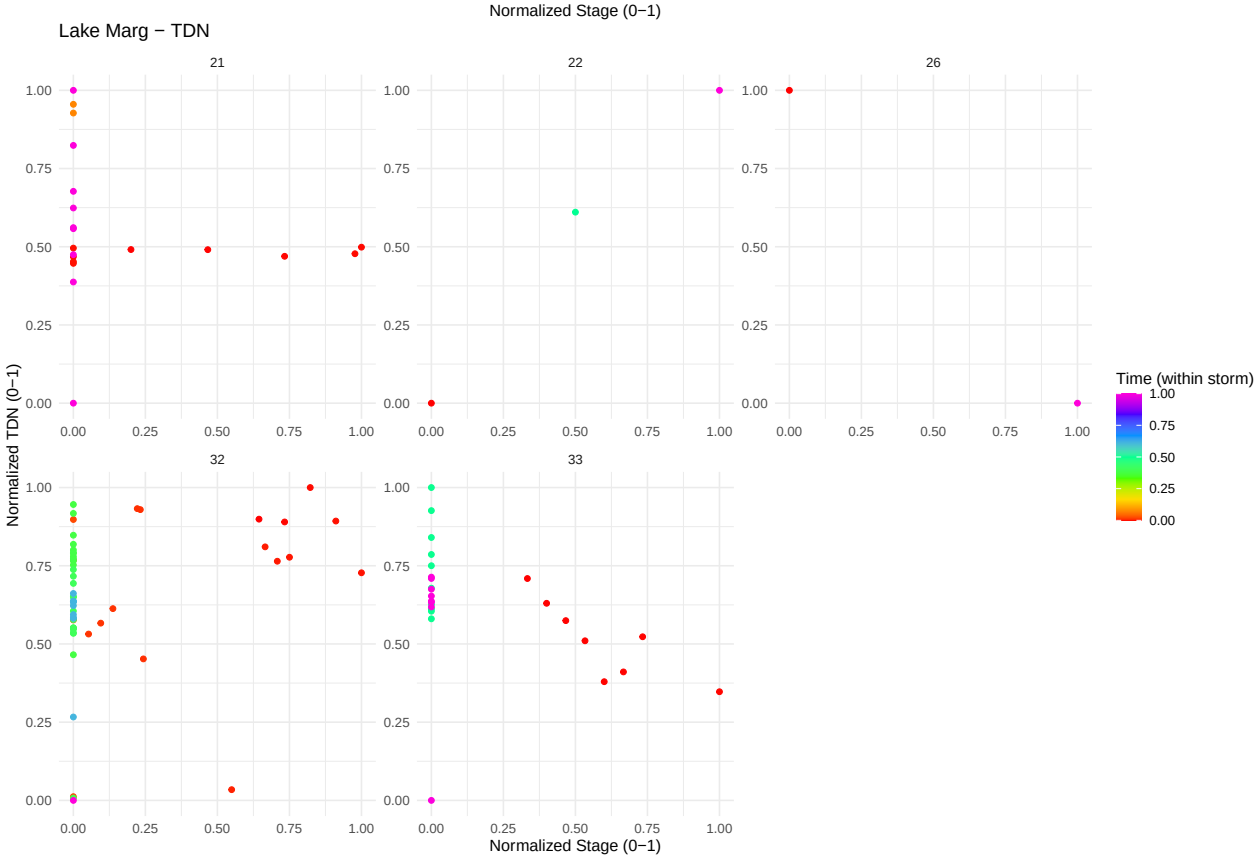

Lake Marg − NH4
21
22
26
32
33
Normalized NH4 (0−1)
Normalized Stage (0−1)
Time (within storm)
0.00
0.25
0.50
0.75
1.00

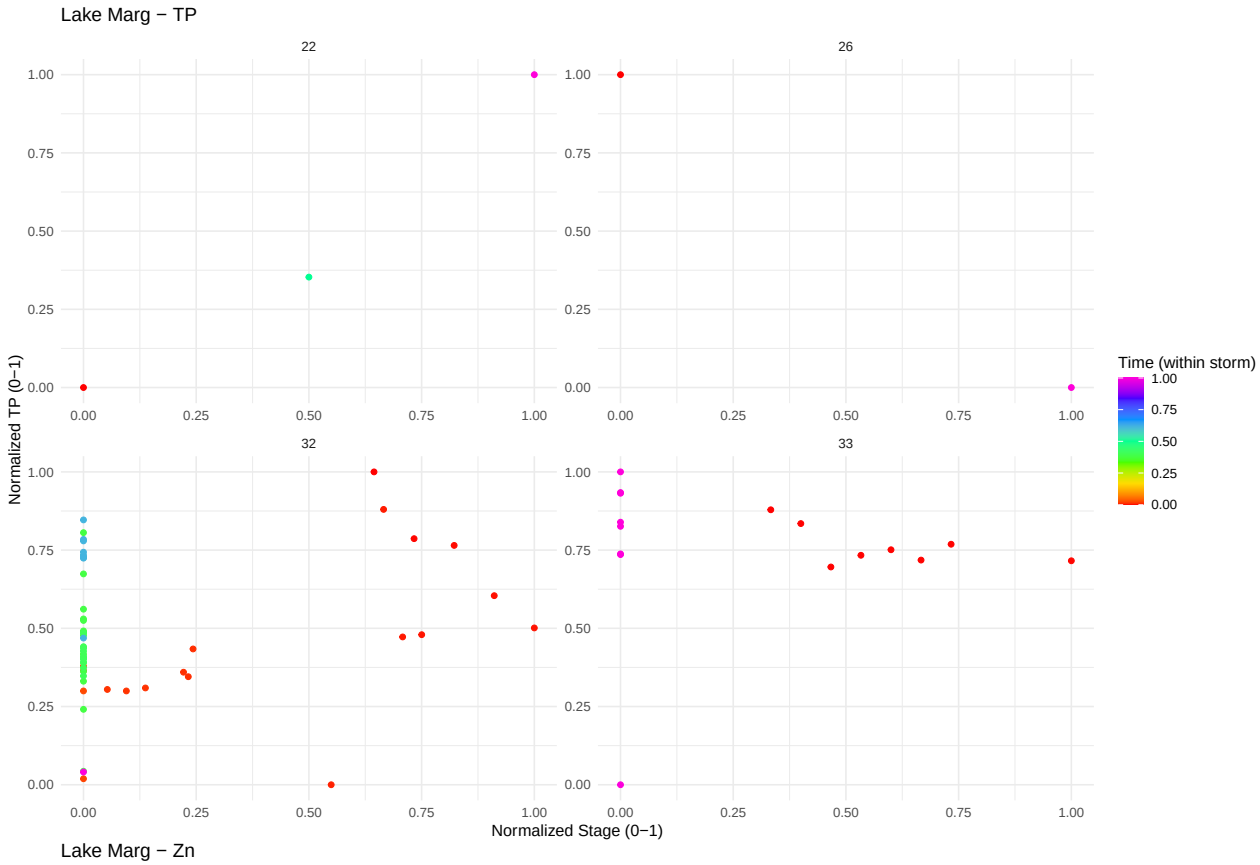

Lake Marg − NO2
21
22
26
32
33
Normalized NO2 (0−1)
Normalized Stage (0−1)
Time (within storm)
0.00
0.25
0.50
0.75
1.00

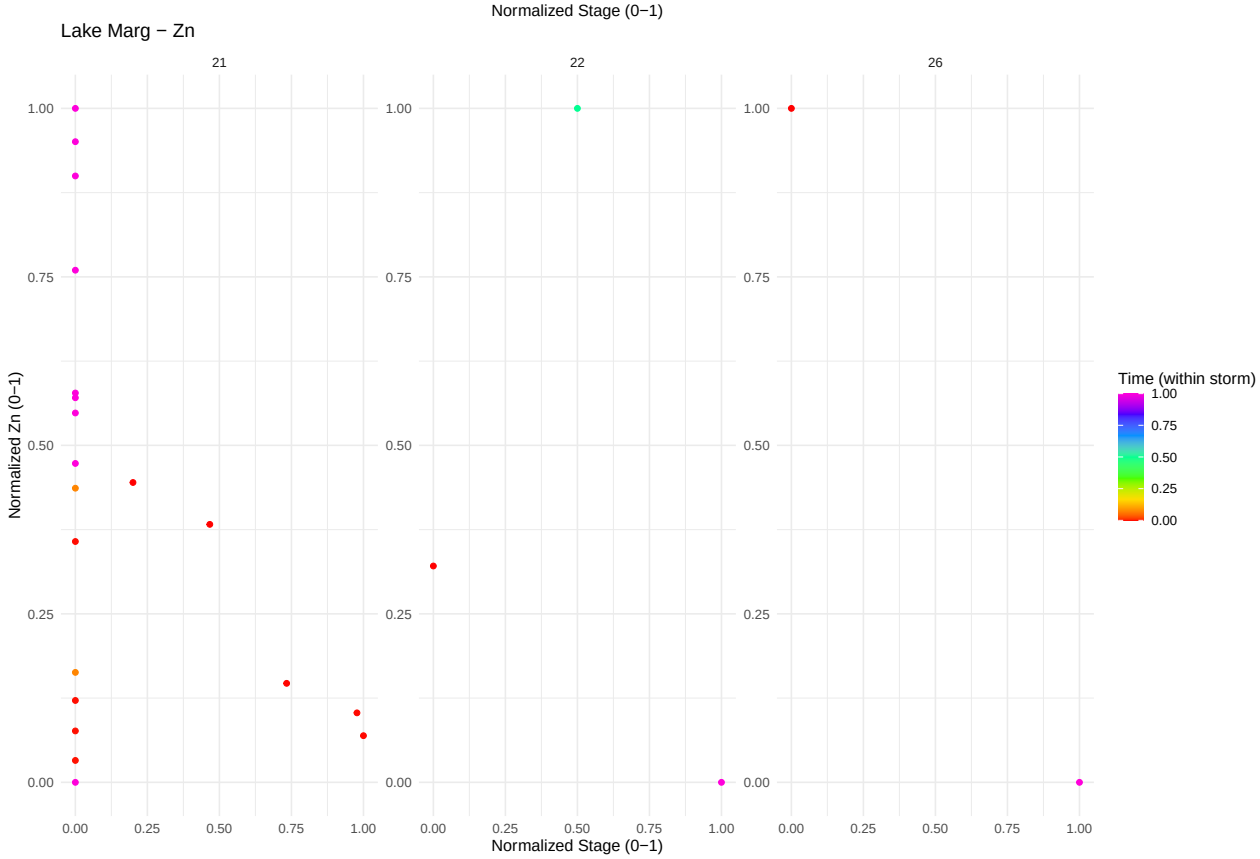

Lake Marg − NO3
21
22
26
32
33
Normalized NO3 (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Lake Marg − Na
21
22
26
Normalized Na (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Lake Marg − SRP
21
22
26
32
33
Normalized SRP (0−1)
Normalized Stage (0−1)
Time (within storm)
0.00
0.25
0.50
0.75
1.00


Lake Marg − TDN
21
22
26
32
33
Normalized TDN (0−1)
Normalized Stage (0−1)
Time (within storm)
0.00
0.25
0.50
0.75
1.00


Lake Marg − TP
22
26
32
33
Normalized TP (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00


Lake Marg − Zn
21
22
26
Normalized Zn (0−1)
Normalized Stage (0−1)
Time (within storm)
1.00
0.75
0.50
0.25
0.00